

PROJECT APPRAISAL

Module 2

The Financial Cash Flow of a Project



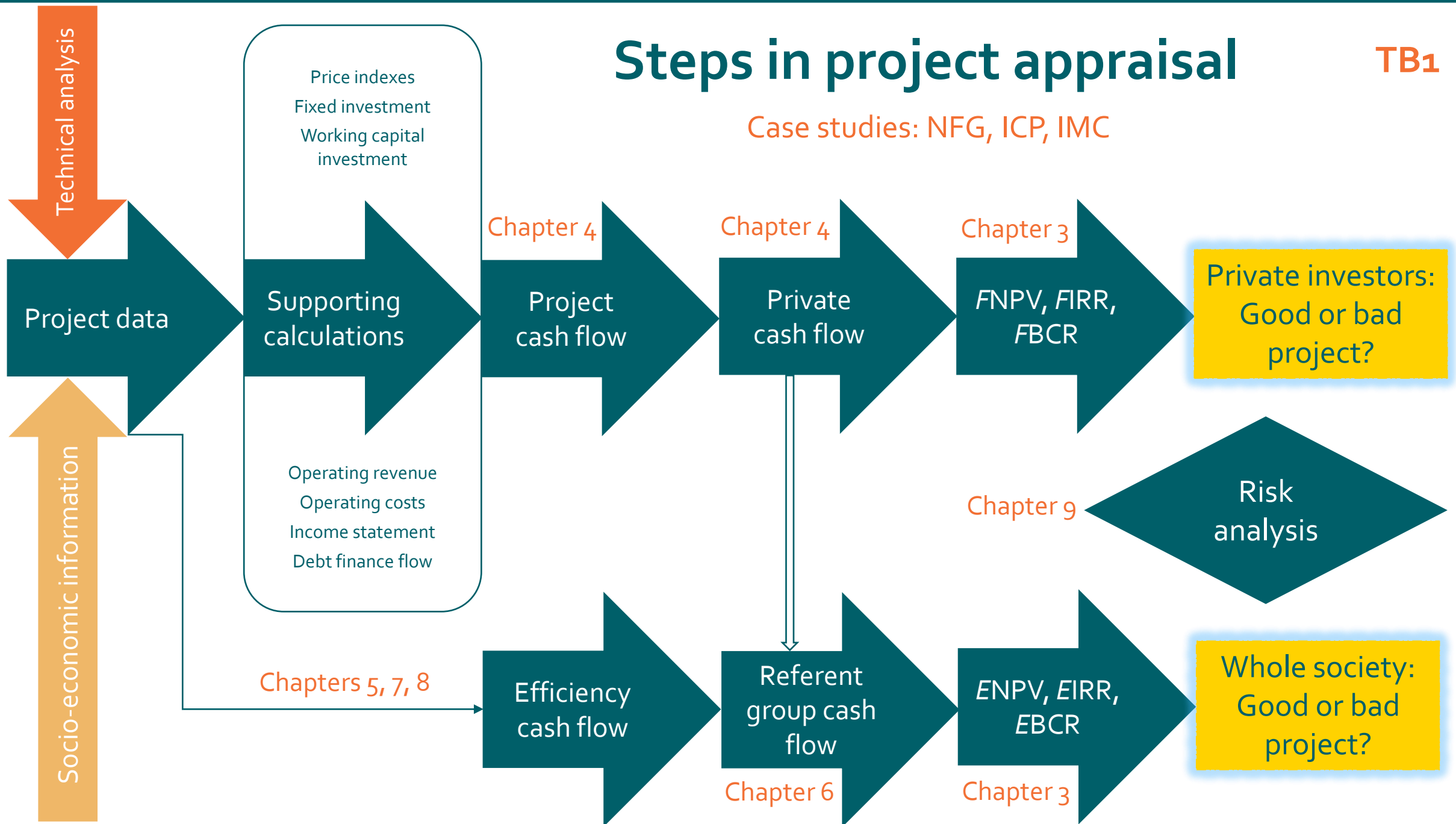
Outline – Module 2

- Introduction to financial analysis of projects
- Financial cash flows: Concepts and characteristics
- Financial cash flows: Principles and conventions
- Financial cash flows: A step-by-step construction
- Concept of incremental cash flows
- An illustrative example

Steps in project appraisal

TB1

Case studies: NFG, ICP, IMC



Financial cash flow of a project

Introduction to financial analysis



What is the financial analysis of a project?

- A project is being evaluated by an enterprise from a purely commercial or “private” perspective, account is taken **only** of the benefits and costs of the project that accrue to the enterprise itself and that affect its profitability.
- If externalities do not affect the enterprise’s financial position, they are omitted from financial analysis.
- The costs and benefits of a project to the firm are values calculated at **market prices**.

What is the financial analysis of a project?

- To calculate the **overall profitability of a project**, we first value all the inputs and outputs at market prices to derive the overall project net cash flows. This calculation is referred as the **Market Analysis/Market CBA/Project Analysis/Project CBA**.
- However, the equity holders of the private firm do not receive all the benefits nor incur all of the costs identified by the Market Analysis.
- The firm's lenders contribute to the initial capital costs and receive a portion of the project's net benefits in the form of interest and capital (principal) repayments.

What is the financial analysis of a project?

- However, ...
 - The market value of project output may include indirect tax, such as a sales tax or value-added tax, and this portion of project benefit accrues to government rather than the firm.
 - Since the Market Analysis values inputs at their tax-inclusive price, the net benefits of the project at market prices do not include indirect taxes levied on inputs and received by the government (b/c of **input tax credit**).
 - The project's earnings will be subject to business/corporate income tax (i.e. a direct tax), received by the government.

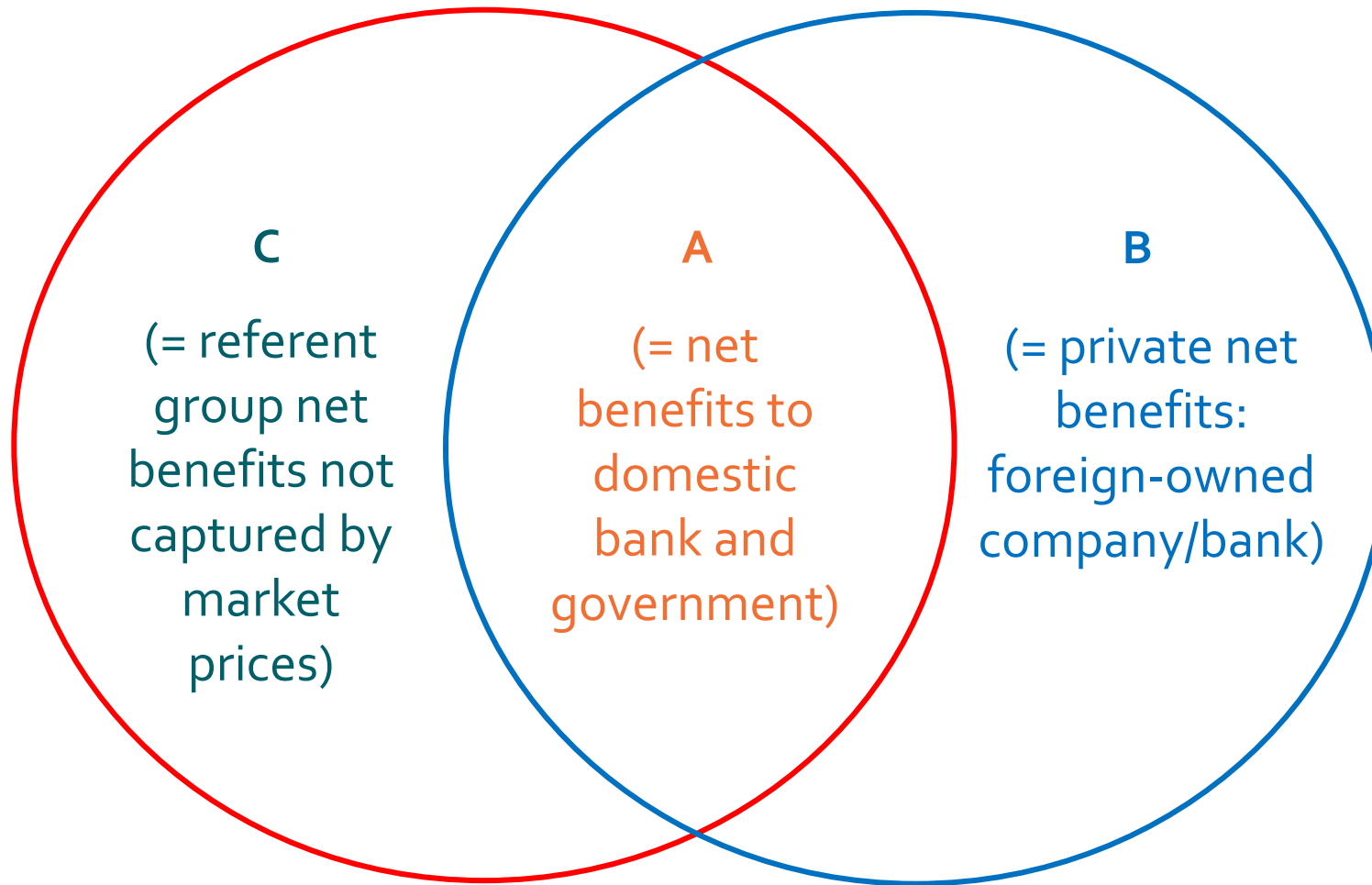
What is the financial analysis of a project?

- Therefore, the net benefit stream identified by the Market Analysis can be adjusted, in Private Analysis, to provide an estimate of **the profitability of the project to its owners** by subtracting debt flows, taxes on output and business income taxes. The **Private Analysis/Private CBA** is widely known as **Equity Point of View (EPV)**.
- By substrating taxes only, we refer the calculation to as after-tax Market Analysis, and it's often named **Total Investment Point of View (TIPV)**. If there is no tax on project output and income tax, TIPV and Market Analysis are identical.

What is the financial analysis of a project?

- The Market Analysis, incorporating the Private Analysis may be important in helping the public decision-maker decide what **incentives** are required in the form of tax concessions.
- In general, the financial analysis of a project helps determine the financial sustainability of the project and its overall success. In doing so, the analyst has to:
 - Organize specific data requirements in certain statements (i.e. cash flow statements – **CFS**);
 - Apply certain investment criteria to these statements.

Source: Campbell & Brown (2016)



A: Referent Group
(market prices)

B: Private Analysis
(market prices)

C: Referent Group
(non-market
prices)

A+B:
Market/Project
Analysis (market
prices)

A + B + C: Efficiency/Economic Analysis (market and
non-market prices)

Financial cash flow of a project

Concepts and characteristics



What is a financial cash flow statement?

- Cash flows are a summary presentation of the various costs and benefits expected to accrue over the project's life. They provide the main basis for deciding whether a project is worthwhile or which of the project's options is the best.
- The financial cash flow statement is a profile of the project's **cash receipts** (inflows) and **cash expenditures** (outflows).
- The project's total expenditures are subtracted from its receipts to provide the **net cash flow** from the project.

Here is an illustration of some of the line items that may appear in the financial cash flow of a project.

Financial Receipts:

1. Sales
2. Changes in Account Receivable
3. Residual Values
 - (a) Land
 - (b) Equipment
 - (c) Buildings
4. Total Inflows

Financial Expenditures:

(i) INVESTMENT EXPENDITURES/OPPORTUNITY COSTS

5. New Investment
 - (a) Land
 - (b) Type 1 Equipment
 - (c) Type 2 Equipment
6. Buildings
7. Existing Assets (if any)
 - (a) Land
 - (b) Equipment
8. Buildings

(ii) OPERATING EXPENDITURES

9. Raw Material (1)
10. Raw Material (2)
11. Raw Material (n)
12. Management
13. Skilled Labor
14. Unskilled Labor
15. Maintenance
16. Changes in Account Payable
17. Changes in Cash Balance
18. Total Outflows
19. Net Cash Flow

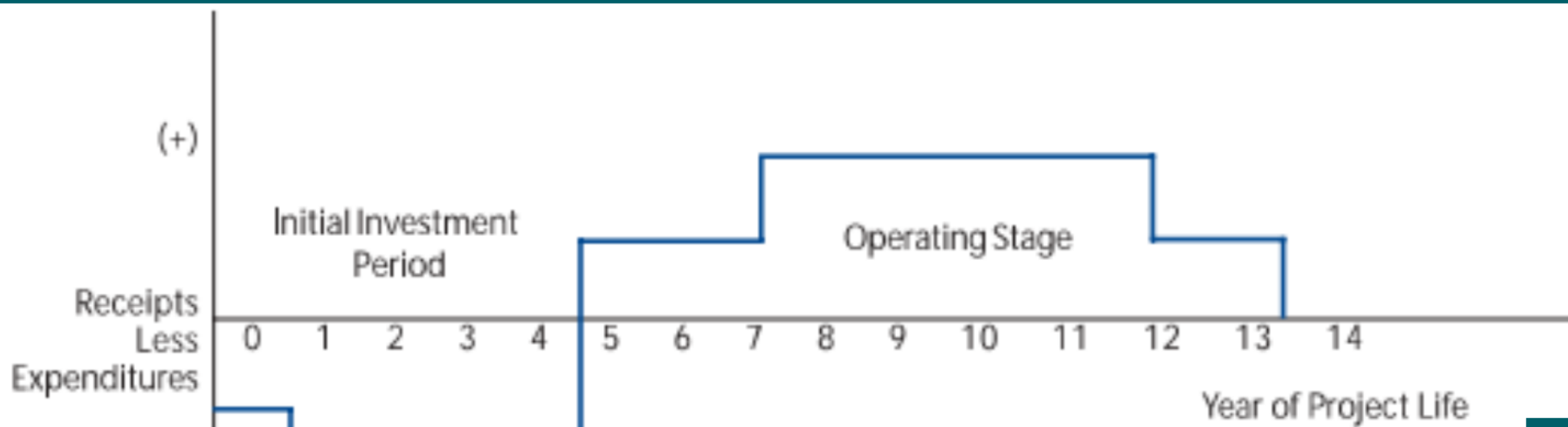
**CASH
INFLOWS**

**CASH
OUTFLOWS**

**NET CASH FLOW
= (4) – (18)**

What is a financial cash flow statement?

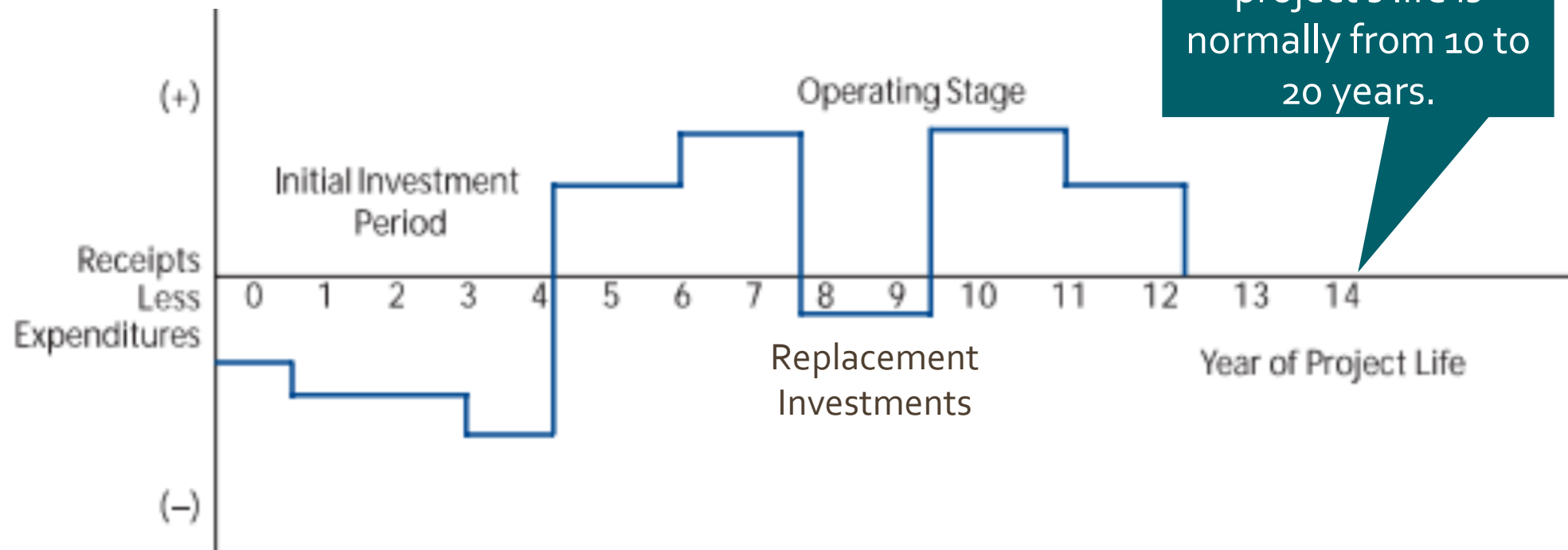
- Typically, a project's net cash flow is negative in the early part of a project's life when the initial investment is being undertaken and the project is not generating any receipts. Once the investment is completed and the project starts operating, the net cash flow is likely to be positive (Panel A).
- There are alternative situations where repalcements, clean-up costs occur (Panels B & C), or the project (does not) generate low (any) financial receipts (Panel D).



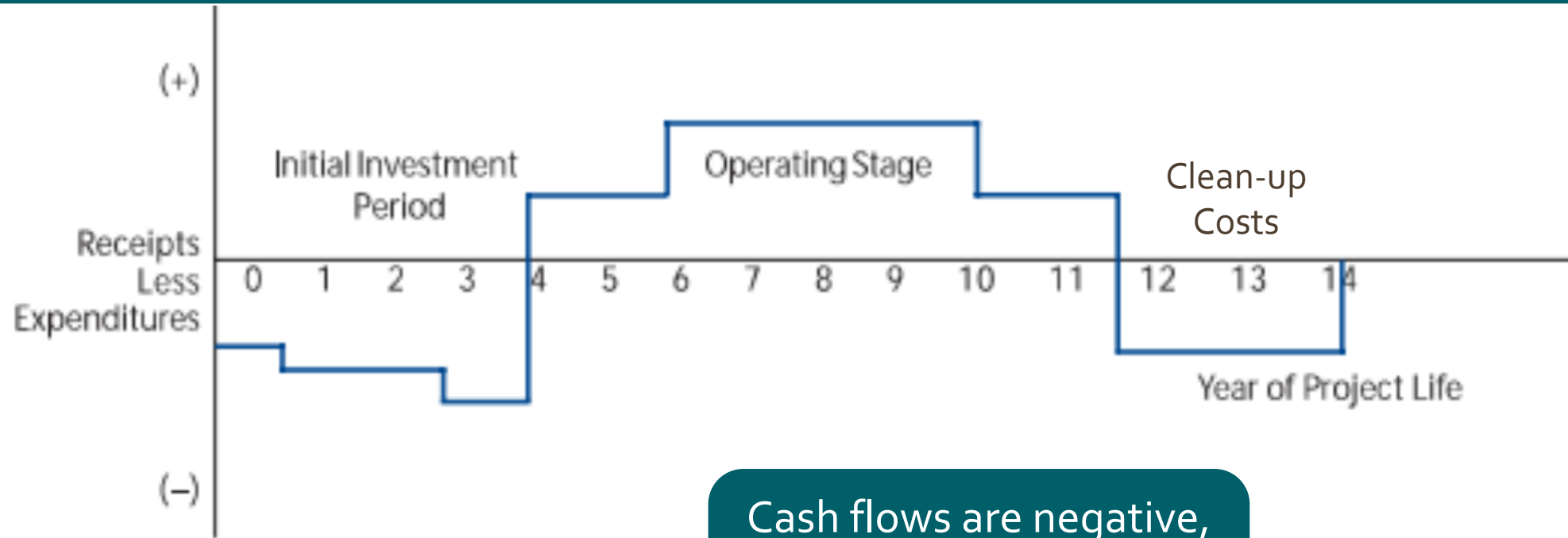
Panel A

The length of a project's life is normally from 10 to 20 years.

Panel B

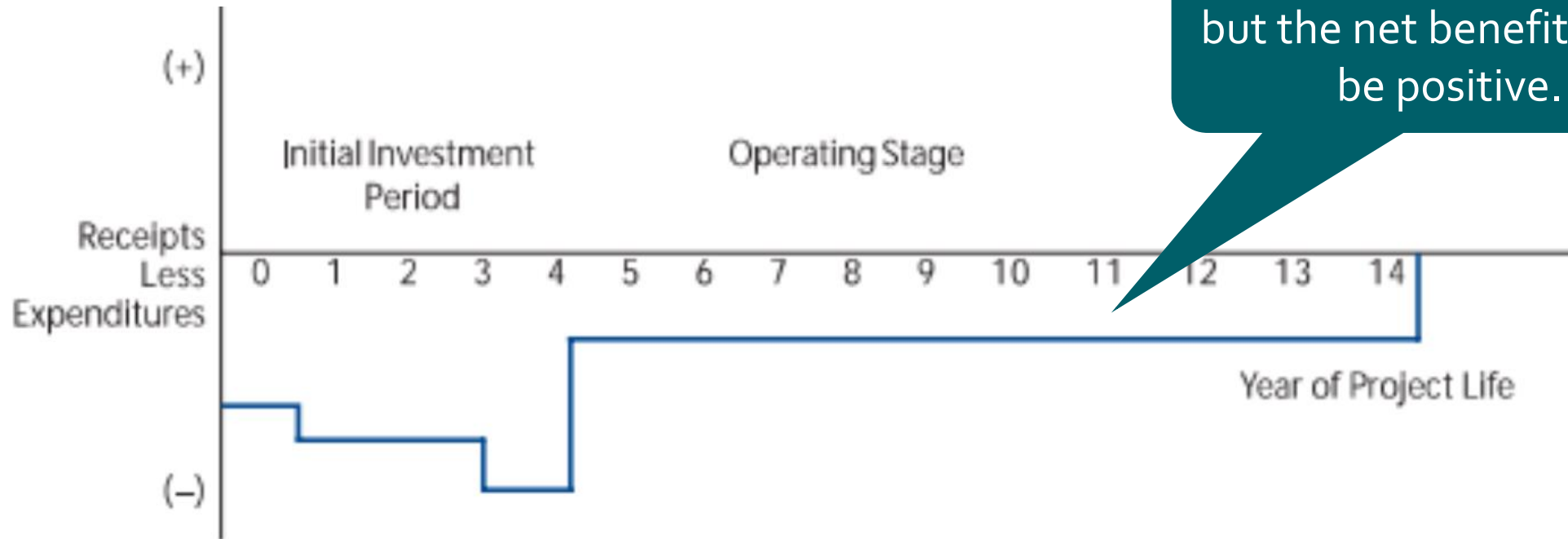


Panel C

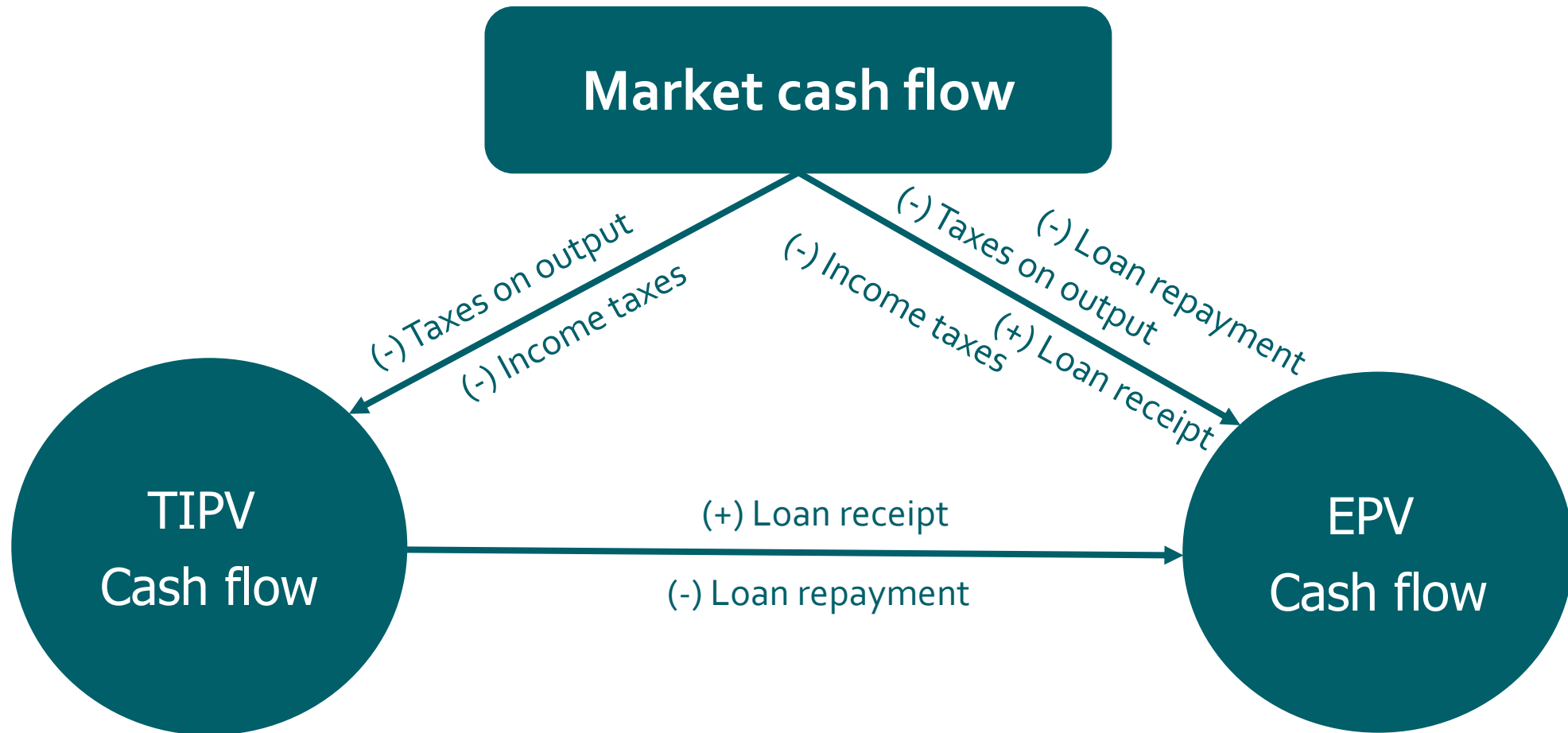


Cash flows are negative, but the net benefits must be positive.

Panel D



What is a financial cash flow statement?



Year	0	1	...	...	n
1. Receipts					
a) Gross sales					
b) Salvage values					
2. Expenditures					
a) Fixed capital investment costs					
b) Changes in working capital					
c) Operating costs					
3. Market net cash flow					

Indirect tax-inclusive prices

These net cash flows are distributed to the equity owners, lenders, and government.

The relative importance of investment and operating costs can vary from one type of project to another.

= (1) – (2)

Year	0	1	...	...	n
1. Receipts					
a) Gross sales					
b) Salvage values					
2. Expenditures					
a) Fixed capital investment costs					
b) Changes in working capital					
c) Operating costs					
3. Market net cash flow			$= (1) - (2)$		
4. Indirect taxes					
5. Income taxes					
6. Market net cash flow (TIPV)			$= (3) - (4) - (5)$		

Include indirect taxes, import duties, but exclude subsidies

These net cash flows are distributed to the equity owners and lenders.

Year	0	1	...	...	n
1. Market net cash flow (TIPV)					
2. Loan receipt					
3. Interest and principal payment					
4. Private net cash flow (EPV)	= (1) + (2) – (3)				

$$NCF_{\text{Market}} > NCF_{\text{Private}}$$

These residual net cash flows are distributed to the equity owners.

Financial cash flow of a project

Principles and conventions



Principles of a project's cash flows

Golden rule 1:

Only cash impacts (i.e. cash transactions, cash receipts or expenditures) are included in the cash flow statement.

Golden rule 2:

Avoid double counting of any asset in the cash outflows such depreciation and interest charges.

Project's cost
and benefits

Project's
costs

Exception 1:

Existing assets are not cash expenditures, but they are taken into account in the form of their opportunity costs.

Exception 2:

Sunk costs were cash expenditures in the past, but being excluded from the cash outflows because they have no alternative use.

Principles of a project's cash flows

Year	0	1	2	3
Sales		1000	1500	
Accounts receivable		100	150	
Cash receipts		900	1450	150

Year	0	1	2	3
Sales		1000	1500	
ΔAR		-100	-50	150
Cash receipts		900	1450	150

$$\begin{aligned}\text{Cash receipt}_2 &= 1450 = 1500 + 100 - 150 \\ &= \text{Sales}_2 + AR_1 - AR_2\end{aligned}$$

Year	0	1	2	3
Purchases		1000	1500	
Accounts payable		200	300	
Cash expenditures		800	1400	300

Year	0	1	2	3
Purchases		1000	1500	
ΔAP		-200	-100	300
Cash expenditures		800	1400	300

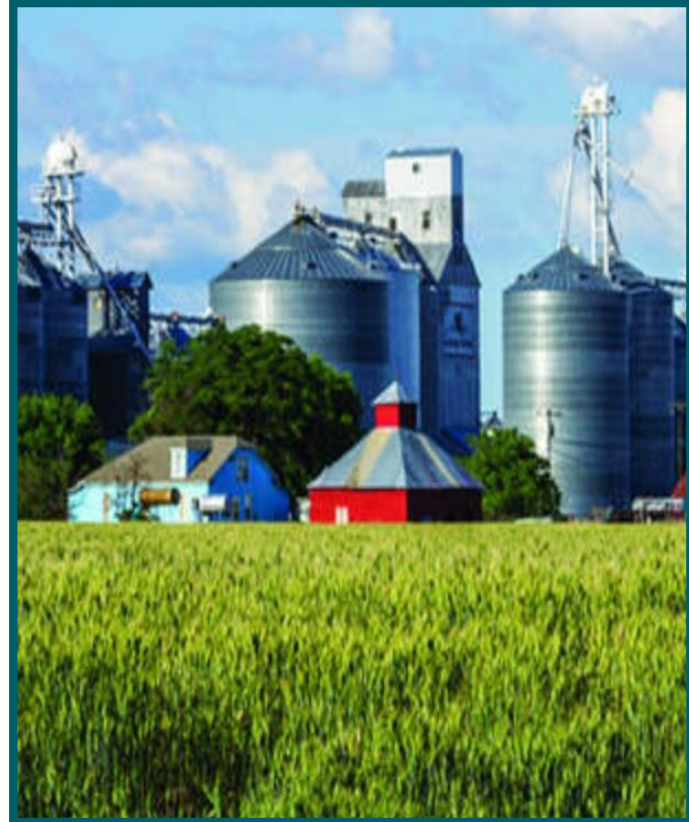
$$\begin{aligned}\text{Cash expenditure}_2 &= 1400 = 1500 + 200 - 300 \\ &= \text{Purchase}_2 + AP_1 - AP_2\end{aligned}$$

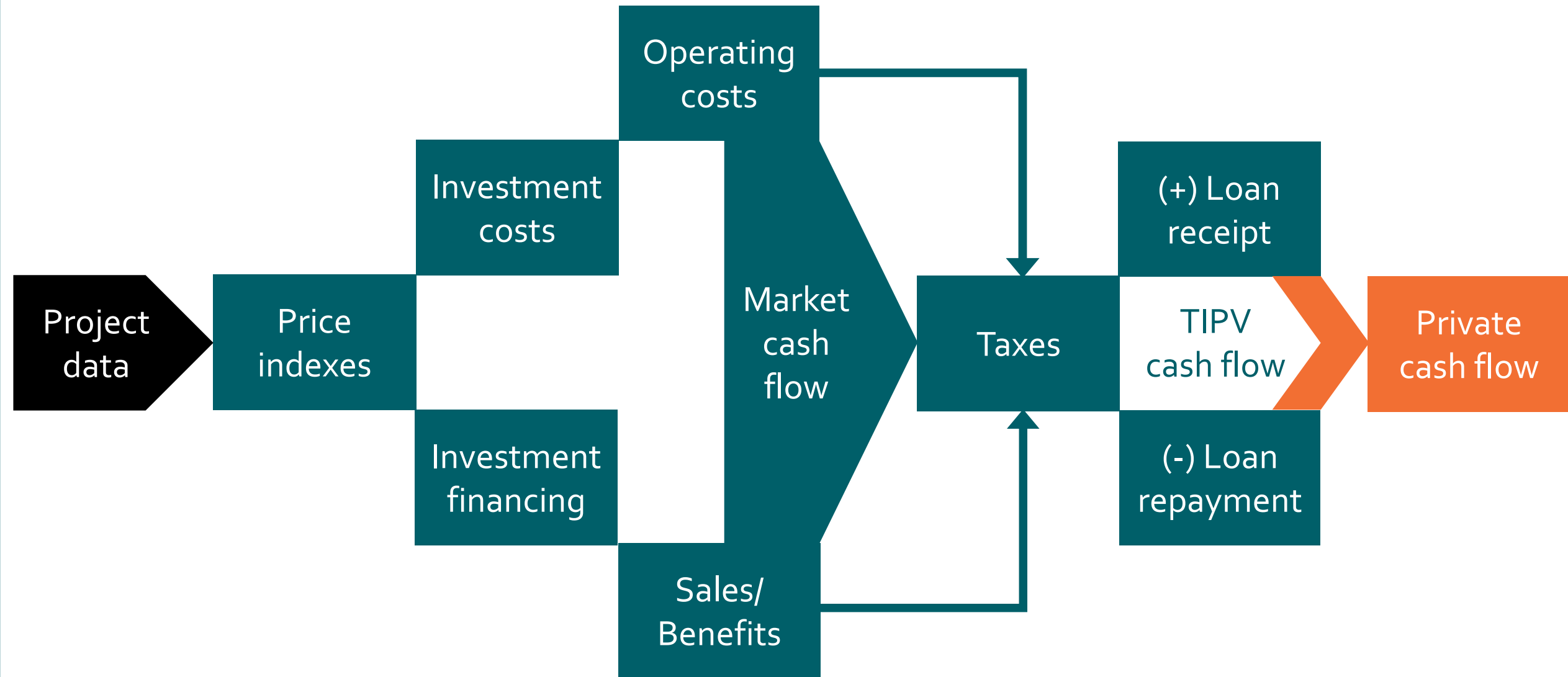
Conventions of a project's cash flows

- We call the total number of years covered by a project's cash flows as the length of a **project's life** or “lifetime”, or sometimes “time horizon”.
- The project's life depends on the *nature of the project* and the *useful life of its main assets*.
- Project years: Year 0, Year 1, ... where **Year 0 is the current**.
- All costs and benefits are concentrated at the end of each project year.
- Unit of currency: Domestic currency
- Depreciation method: **Straight line** (not DDB or any accelerated)
- The market prices of project outputs and inputs include any indirect taxes such as sales tax, value-added tax, excise tax, or payroll tax or tariff on imported goods.

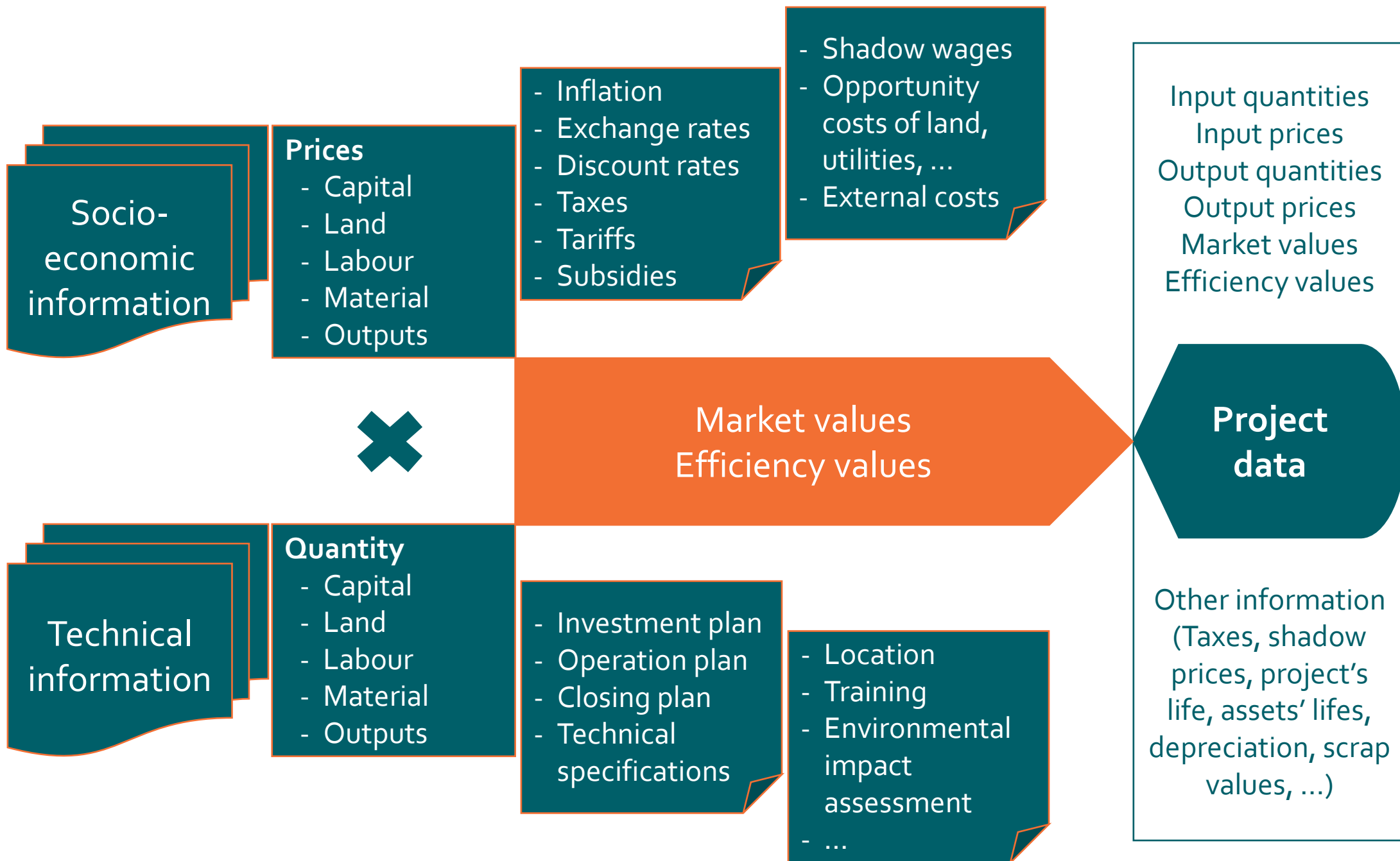
Financial cash flow of a project

A step-by-step construction





Project data (Introduction)



Project data (Practical guideline)

- The principles of project data preparation:
 - Any project value of a variable should be manually **entered** into the spreadsheet only once.
 - Any subsequent use of that data should be in the form of a reference to the address of the cell into which the value was first entered.
 - This means **any change in the project data will automatically result in change in the project outcomes**.
- Suggestions for a single spreadsheet's calculations:
 - Constant data throughout the project's life is manually entered in column dimension at the Year 0 position.
 - Variable data such as capacities, loan disbursement, etc., are manually entered in row dimension at the corresponding years.
 - Units of analysis must be clearly defined and described. A **conversion factor** such as 1.000, 1.000.000 can be used.
- Popular order of the project data:
 - Macroeconomic information
 - Investment costs (initial investments, replacement investments, working capital; quantities, market prices, efficiency prices, investment values, depreciation, and the scrap values)
 - Capital structure, financing information, repayment term, and repayment method
 - Sales related data (quantities, market prices, efficiency prices, output values, "account receivables", ...)
 - Operating costs data (fixed and variable operating costs, market prices, efficiency prices, cost values, "account payables", "cash balance", ...)
 - Taxes, tariffs, shadow prices, discount rates, etc.

Project data (Examples)

NFG Project (Campbell & Brown, 2016)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1																			
2		Table 1: Key Variables				Project		Efficiency					Project		Efficiency Pricing				
3		Investment Costs	No.	Price	Cost	Price	Cost					No.	Price	Cost			%	Price	Cost
4		(i) Fixed Investment	(units)	(\$)		(\$)			Rent on land (Ha)	100	30	3,000	Rent op. cost	0%	0.00	0			
5		Farm equipment (units)	4	100,000	400,000	90,909	363,636		Fuel (litres)	2,500	0.7	1,750	Fuel tax	10%	0.64	1,591			
6		Vehicles (units)	3	30,000	90,000	25,000	75,000		Seeds (Kg)	250	20	5,000	Seed subsidy	25%	26.67	6,667			
7		Buildings (m2)	250	1,000	250,000	1,000	250,000		Fertilizers (tonnes)	3	500	1,500	Fertilizer Subsidy	30%	714.29	2,143			
8		TOTAL			\$740,000		\$688,636		Insecticides (litres)	3,000	30	90,000	Insecticide subsidy	20%	37.50	112,500			
9		(ii) Working Capital							Water (ML)	900	20	18,000	Water subsidy	40%	33.33	30,000			
10		Fertilizer Stocks (tons)	2	500	1,000	714.29	1,429		Spares	12	1,000	12,000	Spares -duty	15%	869.57	10,435			
11		Insecticide stocks (Litre)	2500	30	75,000	37.50	93,750		Casual labor (days)	100	60	6,000	Labour op. cost	20%	12.00	1,200			
12		Equipment spare parts (	10	1,000	10,000	870	8,696		Administration (/month)	12	1000	12,000	Administration (/month)			12,000			
13		Fuel stocks (Litres)	500	0.7	350	0.64	318		Insurance (p.a.)	1	8,263.5	8,263	Insurance (p.a.)			8,263.5	8,263		
14		TOTAL			\$86,350.0		\$104,192		Management (/month)	12	3,000	36,000	Management			3,000	36,000		
15		(iii) Salvage Value	10%		\$74,000		\$68,864		Miscellaneous	1	7,700	7,700	Miscellaneous			7,700	7,700		
16									TOTAL (Market prices)			\$201,213	TOTAL (Efficiency Prices)			\$228,499			
17		Depreciation	Life(yrs)	Amount p.a.															
18		Equipmnt	10	40,000					Apples (tons)	100	1,000	100,000	External Costs			\$10,000			
19		Vehicles	5	18,000					Peaches (tons)	90	1,250	112,500							
20		Buildings	20	12,500					Pears (tons)	75	1,500	112,500	Import duties						
21		Financing	Amount	Interest	Life (yrs)				TOTAL			\$325,000	Equipment	10%					
22		Loan	\$700,000	3.5%	10								Vehicles	20%					
23		Overdraft	\$40,000	5.0%	4				Capacity Output										
24		Discount rate =	5.0%	10.0%	15.0%				Year	2001	2002	2003	2004+			Conversion factor	1000		
25		Tax rate on profits =	25.0%						%	25%	50%	75%	100%						

Project data (Examples)

ICP Project (Campbell & Brown, 2016)

	A	B	C	D	E	F	G	H
1	OPTION 1 (BANGKOK - NO CONCESSIONS): KEY VARIABLES							
2								
3	Exchange Rate Bt/1US\$	44						
4	ICP Capacity (lbs of yarn)	15,000,000						
5	ICP demand for yarn (lbs)	10,000,000						
6	ICP yarn for sale (lbs)	5,000,000						
7								
8		Capacity Output		Discount Rates				
9	1999	nil		1	4%			
10	2000	50%		2	8%			
11	2001-2009	100%		3	12%			
12				4	16%			
13				5	20%			
14								
15	REVENUES							
16		Rates	Cost Savings	Yarn Sales				
17	CIF price of yarn/lb (\$US)	0.58						
18	CIF price of yarn/lb (Bt)	25.52	255,200,000	127,600,000				
19	Import Duty/lb yarn (Bt)	2.10	21,000,000	10,500,000				
20	Cstms Charges/lb yarn (Bt)	0.30	3,000,000	1,500,000				
21	TOTAL		279,200,000	139,600,000				

Students are asked to enter these projects's data into Excel spreadsheets for coming practical exercises.

ICP Project (Campbell & Brown, 2016)

23	INVESTMENT COSTS: BANGKOK							
24			No. (units)	Cost (US\$)	Cost (Bt)	Import Duty	Total Cost (Bt)	Shadow Price
25	<i>Fixed Investment</i>	<i>[1999]</i>						
26	Spindles		40,000	4,000,000	176,000,000	5%	184,800,000	176,000,000
27	Ancillary equipment			850,000	37,400,000	5%	39,270,000	37,400,000
28	Installation	(skilled)		100,000	4,400,000		4,400,000	4,400,000
29	Building Construction	(unskilled)		818,182	36,000,000		36,000,000	18,000,000
30	Start-up Costs	(skilled)		409,091	18,000,000		18,000,000	18,000,000
31	TOTAL						282,470,000	253,800,000
32								
33	<i>Working Capital</i>	<i>[2000]</i>						
34	3 months' raw materials			1,200,000	52,800,000	10%	58,080,000	52,800,000
35	1 year's spare parts			243,000	10,692,000	5%	11,226,600	10,692,000
36	TOTAL						69,306,600	63,492,000
66	FINANCING							
67			Amount (Bt)	Interest (p/a)	Term (yrs)	Interest on O'Draft		
68	Loan for Spindles & Equip	<i>foreign</i>	160,050,000	8%	5			
69	Working Capital Overdraft	<i>local</i>	60,000,000	10%	9	-6,000,000		
70								
71	TAXES & DEPRECIATION							
72			Rate (%)	Initial Fxd Inv Costs	Dep'n Expense			
73	Profits Tax		50%					
74	Depreciation	<i>[2000 on]</i>	10%	264,470,000	26,447,000			

ICP Project (Campbell & Brown, 2016)

40	OPERATING COSTS: BANGKOK					
	<i>Note: water, power and raw materials cost 50% in 2000, with all other costs 100%.</i>					
41			No.	Cost (US\$)	Cost (Bt)	Import Duty
42	Raw Materials	<i>[2000 on]</i>		0.30		
43	Imported Cotton (lbs)		16,200,000	4,860,000	213,840,000	10%
44						
45			No.	Cost p.a (Bt)	Total Cost (Bt)	Bangkok Shadow Price
46	Labour	<i>[2000 on]</i>	(employees)	(/employee)		50%
47	Supervisors & Technicians		100	150,000	15,000,000	15,000,000
48	Other Skilled Workers		1,000	22,500	22,500,000	22,500,000
49	Unskilled Workers		500	12,000	6,000,000	3,000,000
50	TOTAL		1,600		43,500,000	40,500,000
51						
52	Fuel, Water, Spare Parts	<i>[2000 on]</i>	(lbs of yarn)	(/million lbs)		50%
53	Water		15,000,000	480,000	7,200,000	7,200,000
54	Electricity		15,000,000	480,000	7,200,000	3,600,000
55	TOTAL				14,400,000	10,800,000
56						
57	Spare Parts (% of equipment + 5% duty)	<i>[2000 on]</i>		10%	22,407,000	21,340,000
58	TOTAL				22,407,000	21,340,000
59						
60	Insurance and Rent					
61	Insurance	<i>[2000 on]</i>			3,000,000	0
62	Rent	<i>[2000 on]</i>			3,000,000	3,000,000
63	TOTAL				6,000,000	3,000,000

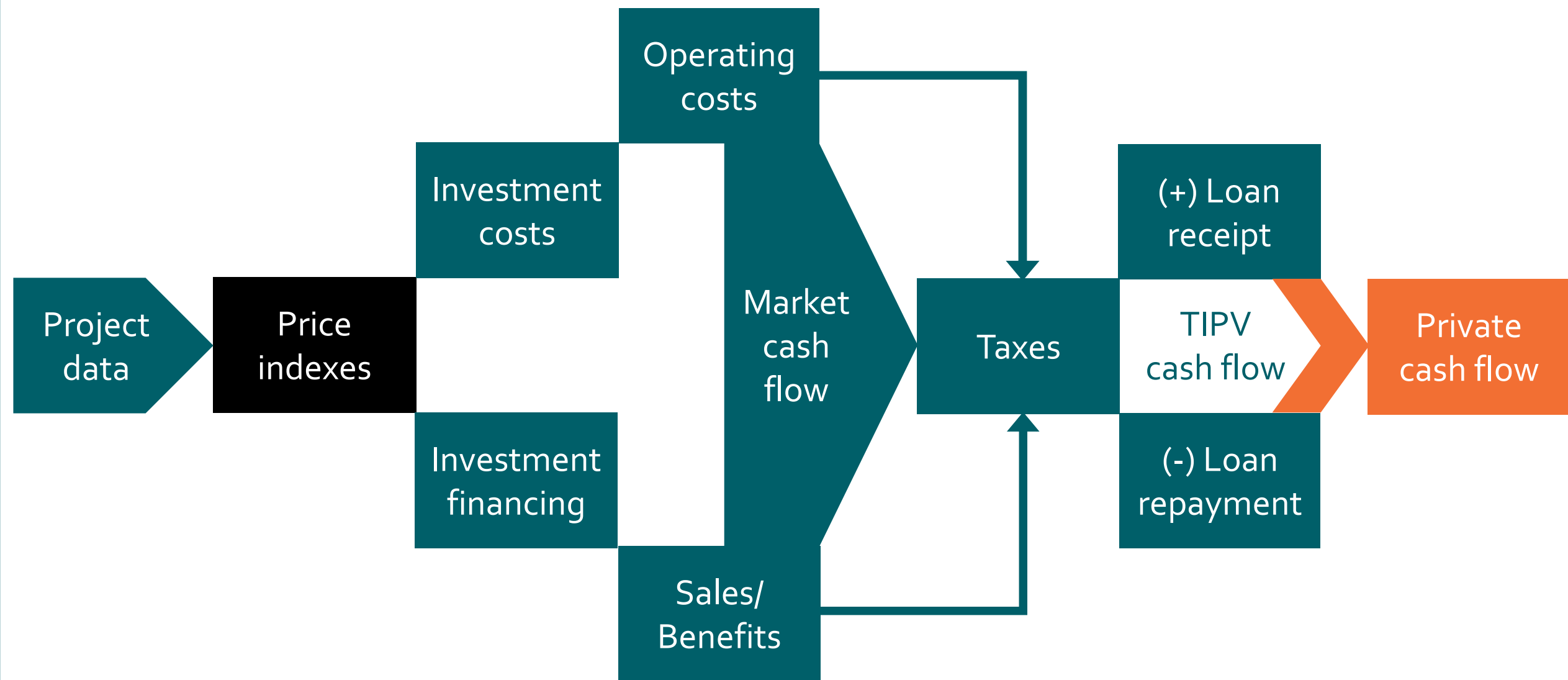
Project data (Examples)

VOI Project (Campbell & Brown, 2016)

A	B	C	D	E	F	G	H	I	J	K	L
1											
2	Units				Prices			Values			
3	Fruit yield Kg/Tree	36			Market	Efficiency		Market	Efficiency		
4	Trees/Ha	300		Land \$/HA	1,000	1,000	-	Land	125,000	125,000	-
5	Oil Yiled litre/kg	0.2		Oil price \$/L	8	6.67	shadow	Trees	480,000	480,000	-
6	Total Land Area Ha	125		Cost Trees \$/Tree	16	16	-	Irrig Infrastructure	1,500,000	1,500,000	shadow
7	Land Wastage %	20%		Land preparation \$/Ha	100	92	shadow	Water Licence	100,000	-	shadow
8	Water use ML/Ha/Annum	6		Planting/watering \$/Ha	1,000	680	shadow	Land Preparation	10,000	9,200	shadow
9				Irrigation Infrastrcture \$/Ha	15,000	15,000	-	Planting and Water	100,000	68,000	shadow
10	Asset Life (yrs)			Infrastructure Repair \$/Ha	7,500	7,500	-	Vehicles & Equip	118,000	102,609	shadow
11	Project life	30		Water Licence	100,000	0	shadow	Shed	40,000	34,400	shadow
12	Irrigation Infrastr. life (yrs)	15		Vehicles & Equipment \$	118,000	102,609	shadow	Total Year 0	2,473,000	2,319,209	-
13	Vehicle life (yrs)	10		Shed & Office \$	40,000	34,400	shadow	Factory Bldng \$	22,500	19,350	shadow
14	Equipment life (yrs)	10		Factory Building \$	22,500	19,350	shadow	Press & Harvester \$	630,000	547,826	shadow
15	Shed and office life (yrs)	30		Press & Harvester \$	630,000	547,826	shadow	Total Year 3	652,500	567,176	-
16	Building and machinery	27		Working Capital \$	50,000	46,739	shadow	Working Capital	50,000	46,739	-
17				Operating Costs (fixed)			-	Fixed O&M			shadow
18	Production Capacity (year)			Tree pruning \$pa	40,000	24,000	shadow	Pruning	40,000	24,000	shadow
19	2006	30%		Fertilizer and herbicides \$/Ha/pa	900	720	shadow	Fertilizer etc	90,000	72,000	shadow
20	2007	40%		Management and Admin \$/pa	90,000	90,000	-	Water	300,000	420,000	-
21	2008	60%		Water cost \$/ML	500	700	shadow	Management	90,000	90,000	shadow
22	2009	90%		Operating costs (Variable)			-	Total Fixed	520,000	606,000	-
23	2010	100%		Repairs & Maint \$pa	36,400	35,213	shadow	Total Variable	125,600	103,843	-
24				Harvesting labour \$/Ha	300	180	shadow				shadow
25	Taxes & Duties			Harvesting fuel \$/Ha	150	150	-				-
26	Profits Tax	-33%		Factory Labour \$/Ha	275	220	shadow				shadow
27	Agricultural Levy \$/Lt	0.8		Factory Power \$/Ha	60	30	shadow				-
28	Import duties - inputs	15%		Bottling \$/Ha	7	6.3	shadow				-
29	Import duties - olives	20%		Freight \$/Ha	100	100	-				-
30											-
31											-
32	Loans										-
33	Loan amount \$	2500000									-
34	Term	15									-
35	Interest commercial	10%									-
36	Interest concessional	2%									-
37											-
38											-
39	Shadow Prices										-
40	Unskilled Labour	60%									-
41	Water used	140%									-
42	Water Licence	0%									-
43	Land	100%									-
44	Power	50%									-
45											-
46	Discount Rates	3%									-
47		6%									-
48		9%									-
49		12%									-
50											-
51	Conversion Factor	1000									-
52	Costs Conversion Factor	-1000									-
53	Reverse Flow Conversion	-1									-
54											-
55	Total production (kg)	1,080,000									-
56	Total production (L)	216,000									-

Cost Composition	Skilled Labour	Unskilled Labour	Imported Materials	Local Materials
Land preparation	80%	20%		
Planting and watering	20%	80%		
Shed and factory	35%	35%	0%	30%
Working Capital			50%	50%
Pruning	0%	100%		
Fertilizing		50%		50%
Management	90%			10%
Harvesting	0%	100%		
Factory Labour	50%	50%		
Repairs & Maintenance	50%	0%	25%	25%
Bottling \$/Ha	0%	25%		75%

Depreciation Schedule	Cost	Life Years	\$/pa
Irrigation Infrastructure: initial	1,500,000	15	100,000
Irrigation Infrastructure: repairs	750,000	15	50,000
Shed	40,000	30	1,333
Vehicles & Equipment	118,000	10	11,800
Factory	22,500	27	833
Factory Equipment	630,000	27	23,333
	Yrs 1-3	Yrs 4-15	Yrs 16-30
Total Depreciation	113,133	137,300	87,300



Price indexes (Inflation)

- **Inflation** is an increase in the **general price level**; rates of inflation are the percentage increases in the general price level per annum.
- In appraisal, do we have to forecast inflation that far ahead in order to be able to value our flows of inputs and outputs in a correct way? Absolutely NO.
- As inflation is an increase in the *general* price level, we can generally ignore it for purposes of project appraisal. Why?
- It affects all costs and benefits in the same way, and therefore will not affect the relative returns on different projects in any real sense. Thus, *we do in practice is to use **constant or real**, Year 0, prices to value all the project's inputs and outputs.*
- However, in periods of inflation not all prices increase at the same rate. *Relative* prices changes, where they are significant, should not be ignored in project appraisal and therefore should be incorporated into our cash flow estimates. Note, relative price changes are not the result of inflation, but occur alongside inflation, because of changes in demand, technological developments, exhaustion of natural resources, and deterioration of environmental quality. Again, avoid double accounting!
- As standard practice, what can be done is to apply some form of **risk or sensitivity analysis** after appraising the project with constant prices.

Price indexes (Inflation)

In cases we need to estimate the nominal cash flows of a project (e.g. to prepare the budget for investment costs), the price index (PI_t) can be calculated as:

Year	0	1	2	...
Project data				
Inflation rate		m	m	...
Price index table				
Price indexes	$PI_0 = 1$	$PI_1 = (1+m)^1$	$PI_2 = (1+m)^2$	...

Year	0	1	2
Constant price	P_0	P_0	P_0
Relative price changes		g_1	g_2
Real prices (P^r)	P_0	$P_1^r = P_0(1+g_1)$	$P_2^r = P_1(1+g_2) = P_0(1+g_1)(1+g_2)$
Inflation rates		m_1	m_2
Nominal prices	P_0	$P_1 = P_1^r(1+m_1)$ $= P_0(1+g_1)(1+m_1)$	$P_2 = P_2^r(1+m_1)(1+m_2) = P_0(1+g_1)(1+g_1)$ $(1+m_1)(1+m_2)$

Price indexes (Inflation)

- The most important feature for integrating expectations about the future rate of inflation (m) into the project appraisal is to ensure that such expectations are consistent with the projections of the nominal rate of interest (i).
- Lenders increase the nominal interest rate on the loans they give to compensate for the anticipated loss in the real value of the loan caused by inflation.
- The nominal interest rate i as determined by financial markets is made up of three major components: (1) the real interest rate r , (2) a risk factor R , and (3) a factor $(1 + r + R)m$ which represents the compensation for the expected loss in purchasing power attributable to inflation.

$$i = r + R + (1 + r + R)m$$

- For example, if the real interest rate r is 6%, the future rate of inflation m will be 5%, and there is no risk to the loan, the nominal interest rate i is charged:

$$\begin{aligned} i &= 6\% + 0 + (1 + 6\% + 0) * 5\% \\ &= 11.3\% \end{aligned}$$

Price indexes (Inflation)

For example, if all costs and benefits of a project are affected by the same inflation rates of 5% per annum (i.e. an increase in general price level).

Year	0	1	2	3
1) Net cash flow (current price)	-100	42.0	44.1	46.31
2) Net cash flow (constant price)	-100	40.0	40.0	40.0
Inflation rates		5%	5%	5%

$$\text{IRR (nominal)} = 15.18\% = 9.7\% + 5\% + 9.7\% \times 5\% = 9.7\% + (1 + 9.7\%) \times 5\%$$

$$\text{IRR (real)} = 9.7\%$$

Cost of capital (real) @6%

Cost of capital (nominal) @ 11.3% = 6% + 5% + 6%*5%

$$\text{NPV (@11.3\%, row 1)} = 6.92$$

$$\text{NPV (@6\%, row 2)} = 6.92$$

**The project's
outcomes are
the same in both
calculations.**

Price indexes (Exchange rate)

- A key financial variable in any project using or producing tradable goods is the market rate of exchange (E^M). This market exchange rate is expressed as the number of units of domestic currency required to purchase one unit of foreign exchange (F).
- The market exchange rate is the current (nominal) price of foreign exchange. The market exchange rate needs to be projected over the life of the project. With a real exchange rate at Year 0 is E^R , the nominal exchange rate at time t is expressed as:

$$E^M_t = E^R * P^D_t / P^F_t$$

where: P^D is domestic price index, and P^F is foreign price index.

- We can also use the following formula:

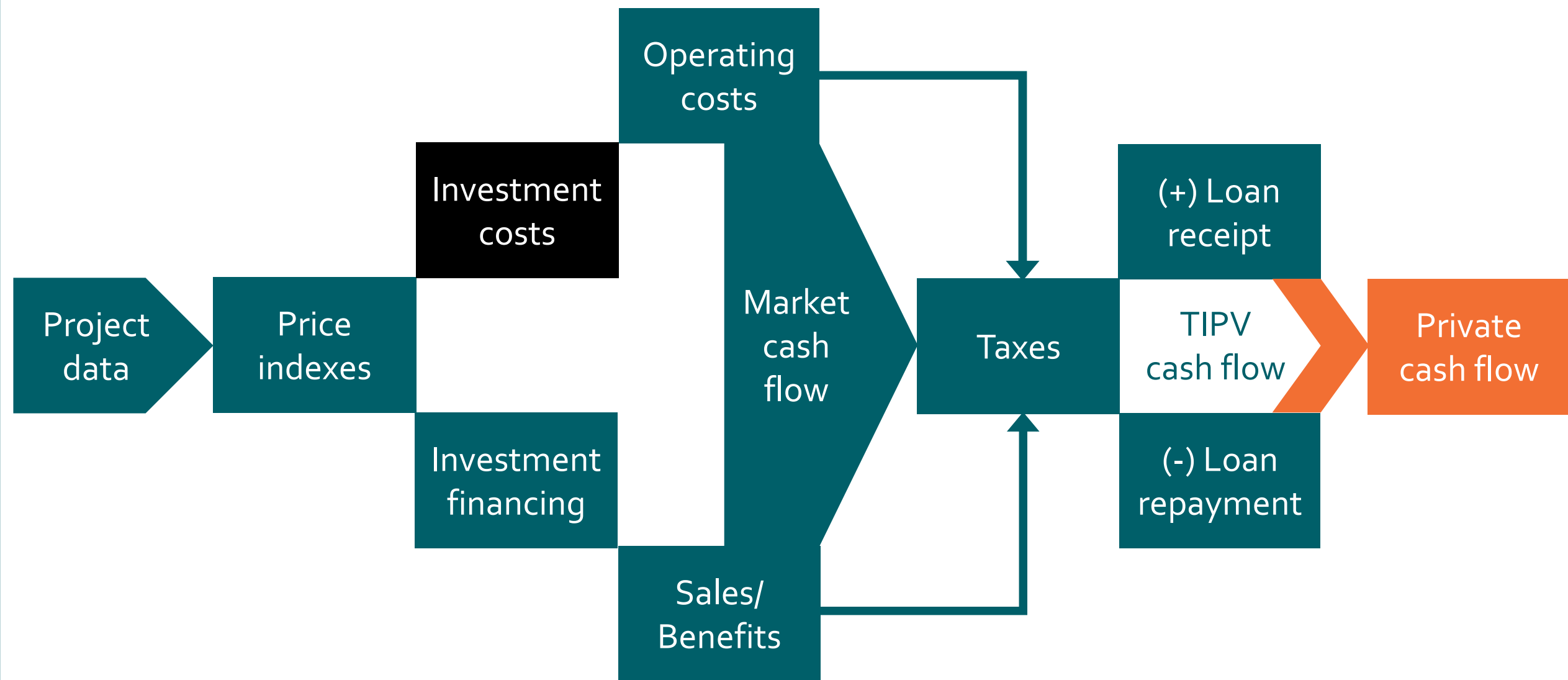
$$E^M_t = E^m_{t-1} (1 + m^D_t) / (1 + m^F_t)$$

Price indexes (Exchange rate)

In Excel spreadsheets, we can set up a relative exchange rate table, along with the price index table.

Year	0	1	2	...
Domestic PI	$PI_0^D = 1$	$PI_1^D = (1+m^D)^1$	$PI_2^D = (1+m^D)^2$	
Foreign PI	$PI_0^F = 1$	$PI_1^F = (1+m^F)^1$	$PI_2^F = (1+m^F)^2$	
Exchange rate	E_0	$E_1 = E_0(PI_1^D/PI_1^F)$	$E_2 = E_0(PI_2^D/PI_2^F)$	

Year	0	1	2	...
Domestic PI	$PI_0^D = 1$	$PI_1^D = (1+m_1^D)$	$PI_2^D = (1+m_1^D)(1+m_2^D)$	
Foreign PI	$PI_0^F = 1$	$PI_1^F = (1+m_1^F)$	$PI_2^F = (1+m_1^F)(1+m_2^F)$	
Exchange rate	E_0	$E_1 = E_0(PI_1^D/PI_1^F)$	$E_2 = E_0(PI_2^D/PI_2^F)$	



Investment costs (Asset treatment)

- **Fixed investment** usually refers to the acquisition of all those capital assets such as land, buildings, plant and machinery that effectively remain intact during the production process, apart from the usual wear-and-tear.
- Investment in fixed capital is usually concentrated in the first year or years of a project's life, but, **replacement investment** will sometimes occur in various years over a project's life.
- At the end of the project period at least part of the fixed investment will be intact: almost certainly the land, buildings, equipment, vehicles, etc. As these can be sold off, it is usual for there to be a positive amount (**cash inflow**) in the last year of the project representing **scrap, salvage or terminal value**.
- In some instances, additional investment-related costs such as **clean-up** costs, rehabilitation costs, etc., could be incurred.

Investment costs (Asset treatment)

- The analyst must deal with the expenditure on new acquisitions and the opportunity cost of existing assets.
 - **New assets** are estimated by using the **Golden Rule 1**.
 - **Existing assets** are estimated by using the **Exception 1**. The opportunity cost of an asset in a specific project is the benefit foregone by not putting the asset to its best alternative use. This cost is generally included in the first year (Year 0) of the project's cash outflow.
- The financial opportunity cost of an existing asset is the highest financial price that could be sold for. The highest financial price is typically the higher of the **in-use value** of the asset and its **liquidation value**.
- The most accurate way to determine in-use and liquidation values is through reliable market assessors. In this case, the assessor's fees should be subtracted from the quoted values.

Investment costs (Data breakdown)

- Once time schedules and deadlines are formulated, expenditures should be broken down **by year** of expected expenditure.
- Expenditures on internationally traded items should be separated from expenditures on items that are not traded internationally.
- Also, other expenditure items should be **broken down into its components**, whenever possible and appropriate.
- For examples, an imported item can be broken down by what suppliers receive, import tariffs, VAT, freight charges, handling charges, installation costs, etc. A building construction can be broken down into raw materials, skilled labour, and unskilled labour.
- These breakdowns are necessary for conducting the efficiency analysis of the project and also important for providing a clear understanding of the project's cost structure.

Investment costs (Data breakdown)

	Year 1			Year 2			Year 3			Year 4		
	Local	Foreign	Total	Local	Foreign	Total	Local	Foreign	Total	Local	Foreign	Total
a. Water Reservoirs/Pumping Stations												
Civil Works	5.3	2.3	7.6	13.0	5.6	18.6	8.8	3.8	12.6	1.2	0.5	1.7
Equipment and Materials	0.5	2.1	2.7	1.6	6.3	7.9	0.7	3.0	3.7	0.2	0.9	1.1
b. Transmission Mains												
Civil Works	1.8	4.2	6.0	2.3	5.4	7.7	1.6	3.8	5.4	0.3	0.7	1.0
Equipment and Materials	2.2	9.0	11.2	2.8	11.4	14.2	2.0	8.1	10.1	0.4	1.5	1.9
c. Secondary/Tertiary Networks												
Civil Works	11.2	3.7	14.9	26.0	8.7	34.7	29.8	9.9	39.7	7.4	2.5	9.9
Equipment and Materials	4.5	17.9	22.3	10.4	41.7	52.1	11.9	47.6	59.5	3.0	11.9	14.9
d. Service Connections												
Civil Works	8.3	1.5	9.7	19.3	3.4	22.7	22.0	3.9	25.9	5.5	1.0	6.5
Equipment and Materials	1.9	7.8	9.7	4.5	18.1	22.7	5.2	20.8	26.0	1.3	5.2	6.5
e. Office Buildings	3.7		3.7	3.7		3.7	0.0					
g. Consulting Services	1.1		1.1	0.5		0.5	0.3		0.3			
h. Land Cost	25.0		25.0									
i. In-house Eng. Services	16.3		16.3	16.3		16.3	12.2		12.2	4.1	0.0	4.1
j. Taxes and Duties	12.4		12.4	25.6		25.6	25.7		25.7	6.3	0.0	6.3
Total	94.2	48.5	142.7	126.1	100.5	226.6	120.4	100.8	221.2	29.7	24.1	53.8

Investment costs (Depreciation treatment)

- **Depreciation** is purely a bookkeeping device – it amounts to setting aside some part of the project's income or benefits each year in order to recoup capital costs and finance capital (investment) expenditure in the future.
- Depreciation is included in the accountant's profit and loss (or income statement) as an item of expenditure. Its main aim is to calculate the project's *taxable profits*.
- However, from the point of view of the project analyst, **depreciation should not be treated as a project cost**.
- If we were to include depreciation as a cost, we would be double counting the project's investment cost and therefore undervaluing the project's net benefit (**Golden Rule 2**).
- Remember that all (new) investment costs should be recorded in the cash outflow in the year in which they were actually incurred (**Golden Rule 1**).

Investment costs (Depreciation treatment)

Year	0	1	...	n
Starting (book) value				
Depreciation				
Accumulated depreciation				
Ending (book) value				

Compare: Straight-line, double declining balance, and sum of the years' digits methods?

Income statement

Year	
$NCF_{TIPV}^{\text{before-tax}}$	
Income tax	
$NCF_{TIPV}^{\text{after-tax}}$	
NPV	

Year	n
Receipts	
...	
Salvage value	

Inflation
adjustment

Investment costs (Interest in construction)

- Interest during construction is an item that is often included as an accounting cost in the construction phase (i.e. as a tax deductible). That means the total investment is an interest-inclusive, and total depreciation is higher.
- In both **Market Analysis** and **TIPV**, the cash flows actually include the debt flow. Therefore, investment costs do not include interest charges during the construction phase.
- In other words, these cash flows are the aggregate cash flows of all project “funders” (equity holders and lenders). Interest payment is exactly the transfer between the funders, and thus it is not an actual expenditure.
- If no interest has been paid by the project, then interest during construction is not cash expenditure and should not be included as expenditure in the cash flow statement of the equity holders (but the interest and capital repayment will be higher in operating phase).
- If interest payments have been made during the period of construction, then there is a cash outflow when the project being examined from the **view point of the equity owner (Private Analysis – EPV)**.

Investment costs (Increased cost contingency)

- Increased investment cost contingency is classified in two categories:
 - Real cost increase contingency (based on experiences from previous projects);
 - Price increase contingency due to effect of inflation.
- In financial appraisal of projects, **real cost increase contingency** is an actual cash expenditure, and thus as a part of investment costs.
- **Price increase contingency** due to inflation just increases the nominal value of investment costs.
 - In nominal cash flow analysis, price increase contingency due to inflation is a cash outflow item. It is incorporated into the nominal costs of investment.
 - In real cash flow analysis, the investment cost does not include the price increase contingency because the real cost of investment remains unchanged.

Investment costs (Increased cost contingency)

	A	B	C	D	E	F	G	H
1	Year	2017	2018	2019	2020	2021	2022	2023
2	Period	0	1	2	3	4	5	6
3	PROJECT DATA							
4	Macroeconomic data							
5	VND inflation rate	5.0%						
6	USD inflation rate	1.5%						
7	VND/USD exchange rate	22,700						
8	Change in VND/USD exchange rate (p.a, next 10 years)	1.0%						
9	Investment costs (million USD, 2017)							
10	Machines, equipments	600.0						
11	Foreign consultancy	10.0						
12	Quantity and real cost increase contingency	30.0						
13	Schedule of investment cost outlays		10%	15%	35%	30%	10%	
14	Investment costs (billion VND, 2017)							
15	Buildings	4,000.0						
16	Project management	100.0						
17	Domestic consultancy	200.0						
18	Site compensation and exploration	500.0						
19	Other costs	300.0						
20	Quantity and real cost increase contingency	400.0						
21	Gestation period	6						
22	Schedule of investment cost outlays	10%	15%	25%	30%	20%	10%	
23	Assets' life (year)	20						
24	Financing							
25	Foreign commercial loan (million USD)	640						
26	Loan disbursement (million USD)		64	96	224	192	64	
27	Nominal interest rate USD (%/p.a)	6%						
28	Real interest rate USD (%/p.a)	4.4%						
29	Grace period (year)	2						
30	Term of loan (year)	10						

Investment costs (Increased cost contingency)

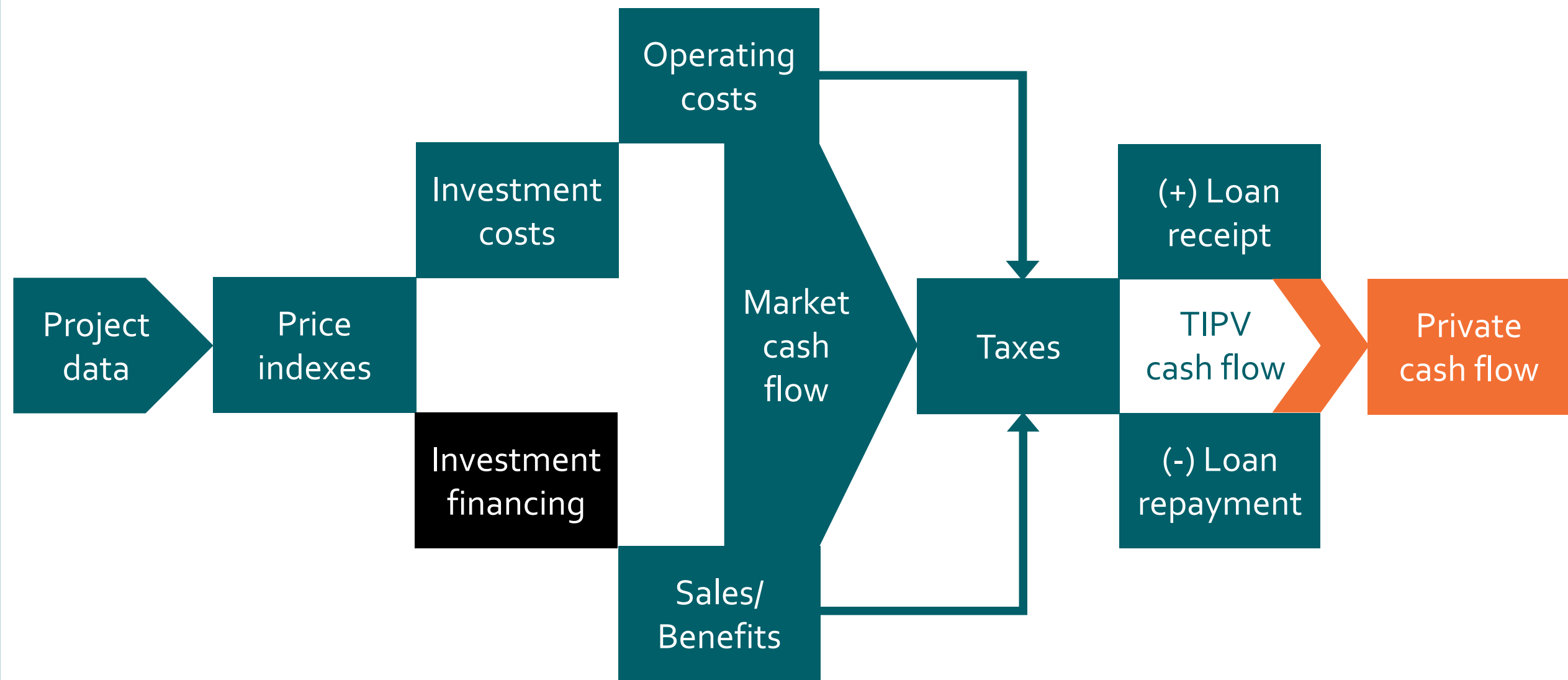
	A	B	C	D	E	F	G	H	I	J
1	Year	2017	2018	2019	2020	2021	2022	2023	2024	2025
2	Period	0	1	2	3	4	5	6	7	8
32	ANALYSIS									
33	Price indexes									
34	VND inflation index	1	1.05	1.10	1.16	1.22	1.28	1.34	1.41	1.48
35	USD inflation index	1	1.02	1.03	1.05	1.06	1.08	1.09	1.11	1.13
36	VND/USD exchange rate	22,700	22,927	23,156	23,388	23,622	23,858	24,097	24,337	24,581
37	Investment schedule									
38	<i>Foreign (million USD)</i>									
39	Total investment costs, 2017	640								
40	Investment flow, 2017		64.0	96.0	224.0	192.0	64.0			
41	Investment flow, nominal prices		65.0	98.9	234.2	203.8	68.9			
42	USD inflation contingency	30.8								
43	<i>Domestic (billion VND)</i>									
44	Total investment costs, 2017	5,500								
45	Total investment costs (USD equivalent, 2017)	242								
46	Investment flow, 2017	550.0	825.0	1375.0	1650.0	1100.0	550.0			
47	Investment flow, nominal prices	550.0	866.3	1515.9	1910.1	1337.1	702.0			
48	Investment flow (USD equivalent, nominal prices)	24.2	37.8	65.5	81.7	56.6	29.4			
49	VND inflation contingency, (USD equivalent)	53								
50	Loan amortization schedule (million USD)									
51	Nominal interest rate USD (%/p.a)	6.0%								
52	Beginning balance		0.0	64.0	163.8	397.7	613.5	714.3	714.3	714.3
53	Disbursement		64.0	96.0	224.0	192.0	64.0	0.0	0.0	0.0
54	Interest payments		0.0	0.0	0.0	0.0	0.0	42.9	42.9	42.9
55	Interest during the construction stage		0.0	3.8	9.8	23.9	36.8	0.0	0.0	0.0
56	Total payments (interest and principal)							42.9	42.9	97.1
57	Principal payments		0.0	0.0	0.0	0.0	0.0	0.0	0.0	54.2
58	Ending balance		64.0	163.8	397.7	613.5	714.3	714.3	714.3	660.1
59	Debt finance flow		64.0	96.0	224.0	192.0	64.0	-42.9	-42.9	-97.1
60	IRR of debt flow	6.0% = lãi vay								
61	Total interest during the construction	74.3								
62	Depreciation schedule (million USD)									
63	Total deductible assets	1,040.3								
64	Starting value							1,040.3	988.3	936.3
65	Depreciation							52.0	52.0	52.0
66	Ending value							988.3	936.3	884.3

Investment costs (Treatment of land)

- Land has an opportunity cost like other asset when it is used by a project. Even if the land is donated to the project by the government, it should be included as part of the investment cost at a value that reflects the market value of land in the project area (**Exception 1**).
- If the project purchases a piece of land, it is treated as a cash expenditure and the **Golden Rule 1** is applied.
- *Land is a very special asset* because it does **not depreciate** under most situations. However, it is fully liquidated at the rate of inflation at the last year of the project.
- Due to improvements in infrastructure, the value of land being used by a project may increase much faster than inflation during the project's life. In such cases, it is important **not to** include the increase in land value that above inflation as part of the liquidation value of the project.
- In most cases, the increase in the liquidation value of land has nothing to do with the project under appraisal.

Investment costs (Treatment of land)

- An exception occurs when the project either improves or causes damage to the land. In such cases, the amount of the land improvement or deterioration should be added to or subtracted from the real value of the land measured at the beginning of the project to determine the liquidation value of the land at the end of the project.
- Alternatively, the opportunity cost of land can be reflected in the cash flow profile of the project by an annual rental charge. This follows the **Golden Rule 1**. Rental charge is treated as part of operating costs in both cash flow and income statements.
- If the annual rental charge approach is used, neither the initial cost of the land nor its final market value should enter into the cash flow profile of the project.



Investment financing (Treatment of loan & interest)

- The sources of finance used whether equity or grants, domestic short term and long term loans, foreign loans, suppliers' credit, concessional loans and other forms of foreign aid should be identified and the disbursement schedules should be formulated.
- When appraising the project from an owner's point of view (EPV), the loan disbursement is an inflow, and the repayment is an outflow.
- However, the analysis of the total invested capital (i.e., Market Analysis and TIPV) is not concerned with the financing but is looking to determine the financial viability of the project to all investors irrespective of the sources and terms of financing.
- The sources of finance are important hint in identifying the referent and non-referent groups of a project.

Investment financing (Treatment of loan & interest)

Item	Year	0	1	2.....7
A. Investment Expenditures				
<i>(a) Site Preparation, Exploration, and Development</i>				
Materials:				
- Traded (cif)		500.0	500.0	
Tariffs @12%		60.0	60.0	
VAT @10%		56.0	56.0	
- Non-traded		400.0	300.0	
VAT @5%		20.0	15.0	
Labor:				
- Skilled		150.0	100.0	
- Unskilled		200.0	250.0	
<i>(b) Equipment</i>				
Traded (cif)		600.0	2,000.0	
Tariffs @10%		60.0	200.0	
VAT @10%		66.0	220.0	
Total Expenditures		2,112.0	3,701.0	
B. Financing				
<i>Equity</i>		2,012.0	1,201.0	
<i>Domestic Loan (short-term)</i>		100.0	500.0	
<i>Foreign Loan (guaranteed by government)</i>		0	2,000.0	

Total investment is broken down into years and different components.

Total financing equals total investment.

Should specify who are the owners? Domestic or foreign firms!

Investment financing (Loan amortization schedule)

Year	0	1	...	m
(1) Beginning balance				
(2) Accrued interest				
(3) Payment	PMT			
♦ Principal	PPMT			
♦ Interest	IPMT			
(4) Ending balance	$= (1) + (2) - (3)$			

Income statement

Year	
$NCF_M^{\text{before-tax}}$	
Income tax	
$NCF_M^{\text{after-tax}}$	
NPV	

Investment financing (Loan amortization schedule)

Year	0	...
(1) Beginning balance		
(2) Accrued		
(3) Payment		
♦ Principal		
♦ Interest		
(4) Ending balance		

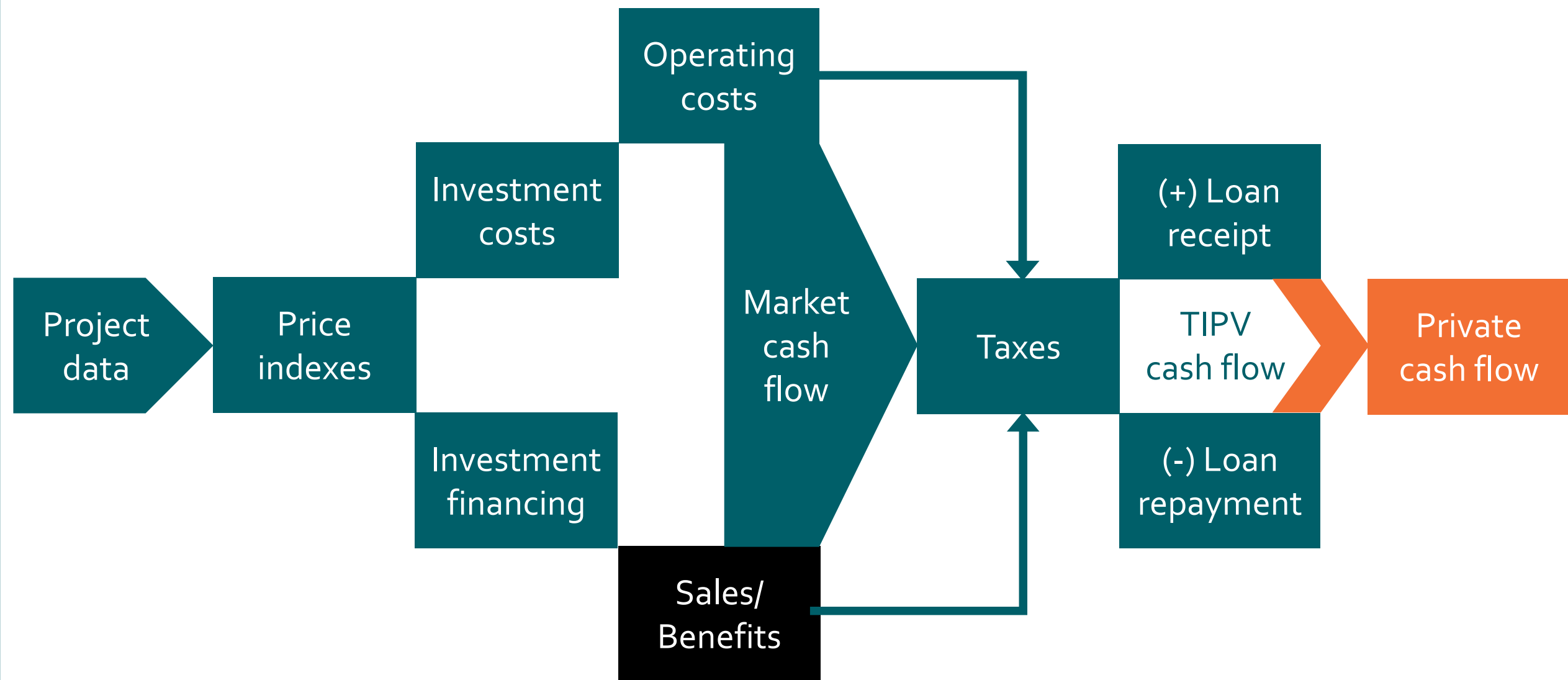
Year	
EBIT	
Interest expense	
EBT	
Income tax	

$NCF_{TIPV, EPV}$

Debt flow

Year	0	1	2	3	4
(1) Loan disbursement					
(2) Total payment					
(3) Debt finance flow	(3) = (1) - (2)				

Year	
NCF_{TIPV}	
NCF_D	
NCF_{EPV}	



Sales/Benefits (Introduction)

- The sales and cash receipts are developed on the basis of the data formulated and organized in the technical and demand (market) modules.
- Project data such as production, **market prices**, sales or value-added taxes, etc. are required to estimate the project's sales, cash receipts and benefits.
- Remember that **gross prices** are used in the cash inflows, the indirect taxes are then adjusted later!
- In cases the project generates goods and services that are subject to government regulations (e.g. ceiling price), the willingness to pay should be estimated in order to construct the **Efficiency Analysis** net benefits.

Sales/Benefits (Adjustment of sales)

- A project's viability is not only determined by the sales it generated but also by the timing of the cash receipts from the sales.
- A cash flow statement records sales transactions only when the cash from the transaction is received (**Golden Rule 1**). Typically, projects forecast their sales as a single line item which comprises both credit and cash transactions. Remember, only cash sales are included in a cash flow statement.
- The cash receipts for any period equals the sales during the period plus the accounts receivable at the beginning of the period less the accounts receivable at the end of the period.
- **Accounts receivable (AR)** are typically measured as a percentage of sales. The appropriate percentage of accounts receivable is up to the current performance of the owner's firm, or should be consistent with the industry standards.

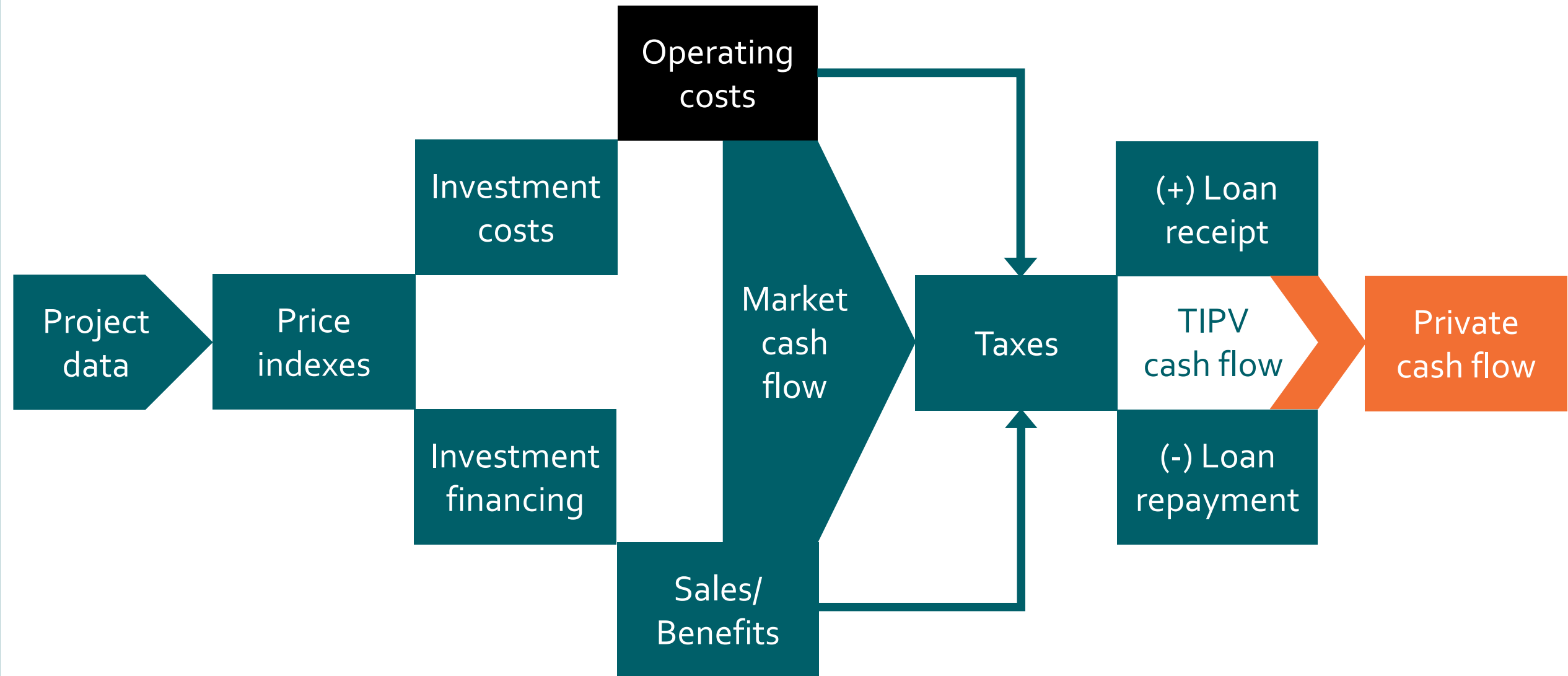
Sales/Benefits (Adjustment of sales)

Year	0	1	2	3	4	5
Sales		1000	1500	2000	1500	
Accounts receivable (10%)		100	150	200	150	
Cash receipts		900	1450	1950	1550	150

Year	0	1	2	3	4	5
Sales		1000	1500	2000	1500	
Changes in accounts receivable		-100	-50	-50	50	150
Cash receipts		900	1450	1950	1550	150

$$\begin{aligned}\text{Cash receipt}_t &= \text{Sales}_t + \text{AR}_{t-1} - \text{AR}_t \\ &= \text{Sales}_t + \Delta \text{AR}_t\end{aligned}$$

$$\begin{aligned}\text{Cash receipt}_2 &= 1450 = 1500 + 100 - 150 \\ &= 1500 - 50\end{aligned}$$



Operating costs (Introduction)

- The purchases and cash expenditure are developed on the basis of the data formulated and organized in the technical and manpower modules.
- Operating costs, i.e. the costs of materials, labour, etc. used to operate and maintain the project. These could be classified into **fixed and variable costs**.
- Project data such as inputs, **market prices**, indirect taxes, import tariffs, subsidies, etc., are required to estimate the project's operating costs.
- Operating expenditures should be broken down into **traded and nontraded** items; and each expenditure item should be broken down into its components.
- These breakdowns are necessary for conducting the efficiency analysis and for providing a better understanding of the cost structure of the operating expenditures.

Operating costs (Adjustment of purchases)

- When the project makes a purchase, the good or service may be delivered to the project but no money transferred from the project to its **suppliers**.
- The cash expenditures for any period equals the purchases during the period plus the accounts payable at the beginning of the period less the accounts payable at the end of the period.
- **Accounts payable (AP)** are typically measured as a percentage of total purchases or that of a major input. The appropriate amount of accounts payable that a project will maintain can be determined on the basis of the current performance of the owner's firm, or should be consistent with the industry standards.

Operating costs (Adjustment of purchases)

Year	0	1	2	3	4	5
Purchases		500	700	800	600	
Accounts payable (10%)		50	70	80	60	
Cash expenditures		450	680	790	620	60

Year	0	1	2	3	4	5
Purchases		500	700	800	600	
Changes in accounts payable		-50	-20	-10	20	60
Cash expenditures		450	680	790	620	60

$$\begin{aligned}\text{Cash expend}_t &= \text{Purchase}_t + \text{AP}_{t-1} - \text{AP}_t \\ &= \text{Purchase}_t + \Delta \text{AP}_t\end{aligned}$$

$$\begin{aligned}\text{Cash expend}_2 &= 680 = 700 + 50 - 70 \\ &= 700 - 20\end{aligned}$$

Operating costs (Adjustment of cash balance)

- Increases and decreases in **cash balances (CB)** owned by the project can take place when non change occurs in sales, **AR**, purchases, or **AP**.
- When cash is set aside for the transactions of the business, it is a use of cash which is represented as an outflow in the cash flow statement.
- A decrease in cash held by the project is a source of cash for the project and its sponsors, and is a cash inflow.
- The amount of cash to be held for facilitating the transactions of the business is typically a percentage of the project's expenditures, sales, or major purchases. It can be determined by examining the performance of similar projects in the same industry.

Operating costs (Adjustment of cash balance)

Year	0	1	2	3	4	5
Sales		1000	1500	2000	1500	
Cash balance (5%)		50	75	100	75	
Changes in cash balance		-50	-25	-25	25	75

$$\Delta CB_t = CB_{t-1} - CB_t$$

Year	0	1	2	3	4	5
Purchases		500	700	800	600	
Cash balance (5%)		25	35	40	30	
Changes in cash balance		25	10	5	-10	-30

$$\Delta CB_t = CB_t - CB_{t-1}$$

Working capital (Introduction)

- Working capital of a project is generally defined as the project's current assets net of its current liabilities.
- Current assets typically include cash and marketable securities, accounts receivable, inventories and prepaid expenses.
- Current liabilities include accounts payable, and any other form of debt that is due within a year.
- **The analysis of working capital has two objectives: (1)** To determine how working capital is accounted for in the cash flow statement, and **(2)** To determine how working capital requirements for a project are estimated and financed.

Working capital (WK in the cash flow)

- Only changes in AR, AP and CB are included in the cash flow statement.
- Changes in prepaid expenses should not be included in the cash flow statement because an expenditure item is recorded as a cash outflow once an actual outlay takes place (**Golden Rule 1**).
- Changes in inventories should not be included in the cash flow statement when a project purchases a certain amount of raw materials, inventories will increase. These inventories are financed through a cash outflow and/or an increase in accounts payable. The **Golden Rule 2** applies for this case.

Working capital (WK in the cash flow)

- The situation is similar when dealing with changes in the inventories of the final product. For example, a decrease in final good inventories implies an increase in sales. This in turn implies an increase in cash receipts or accounts receivable.
- The treatment of working capital in the cash flow statement through its three main components (ΔAR , ΔAP , ΔCB) strictly follows the **Golden Rule 1**: The project's outflows are cash receipts, and the project's inflows are cash expenditures.
- In practice, changes in working capital ($AR + CB - AP$), i.e. working capital at the end of the period less working capital at the beginning of the period, indicates an investment in working capital, and is an outflow's item.

Working capital (WK in the cash flow)

- $\Delta AR_t = AR_{t-1} - AR_t \Rightarrow$ Cash receipts' item
- $\Delta AP_t = AP_{t-1} - AP_t \Rightarrow$ Cash expenditures' item
- $\Delta CB_t = CB_t - CB_{t-1} \Rightarrow$ Cash expenditures' item

Year	0	1	... n
AR	AR_0	AR_1	
AP	AP_0	AP_1	
CB	CB_0	CB_1	
ΔAR		$= AR_0 - AR_1$	
ΔAP		$= AP_0 - AP_1$	
ΔCB		$= CB_1 - CB_0$	

CASH FLOW STATEMENT

Year	
Receipts	
ΔAR	
...	
Expenditures	
...	
ΔAP	
ΔCB	

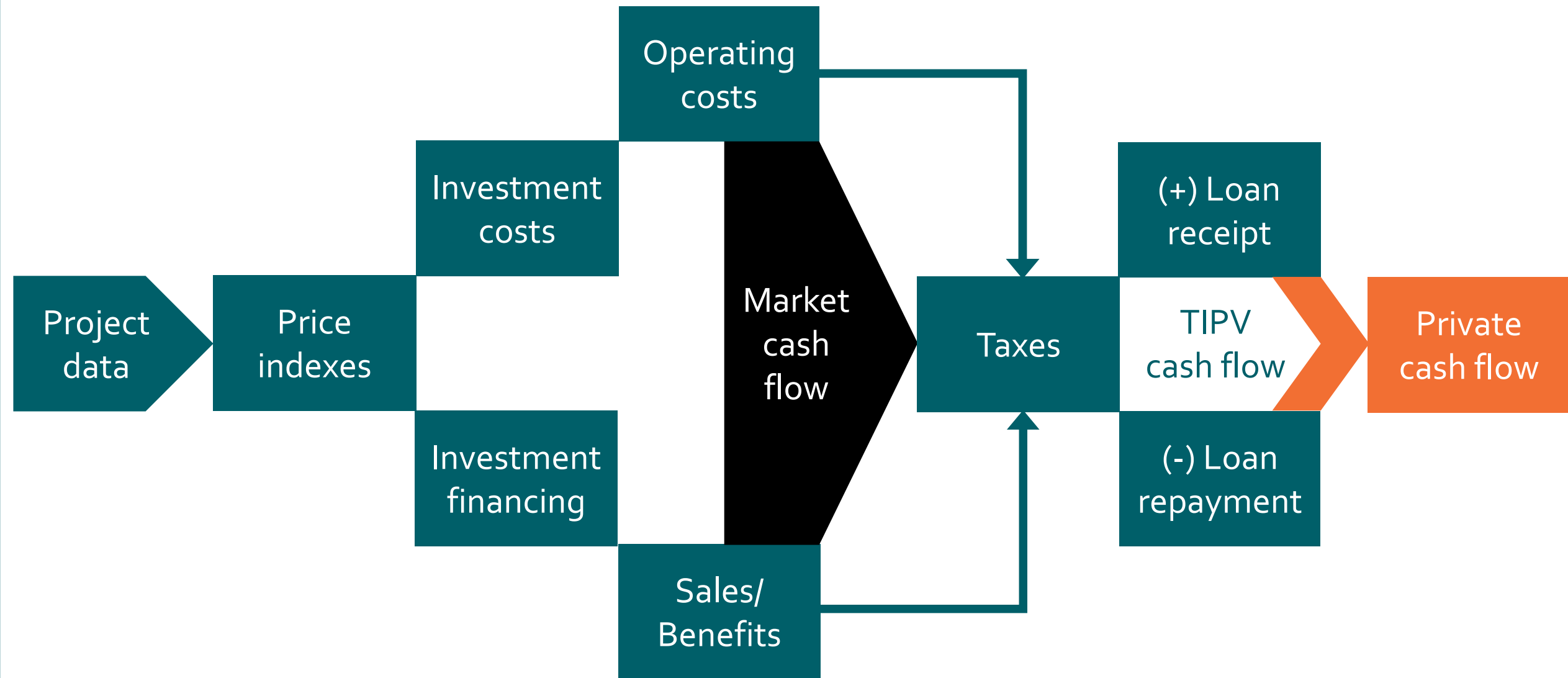
Working capital (WK in the cash flow)

	A	B	C	D	E	F	G	H
1	TWO WAYS OF WORKING CAPITAL TREATMENT							
3								
4	Investment costs	2000						
5	Sales	1000						
6	Sales growth rate	10%	per annum					
7	Operating costs	500						
8	Operating cost growth rate	6%	per annum					
9	AR	15%	Sales					
10	AP	12%	Sales					
11	CB	5%	Sales					
12	Working capital	8%	Sales					
13	Project life	5	years					
14								
15	Year	0	1	2	3	4	5	6
16	Sales		1000	1100	1210	1331	1464	
17	Operating costs		500	530	562	596	631	
18	Accounts receivable		150	165	182	200	220	
19	Accounts payable		120	132	145	160	176	
20	Cash balance		50	55	61	67	73	
21	Working capital		80	88	97	106	117	
22	Changes in accounts receivable (ΔAR)		-150	-15	-17	-18	-20	220
23	Changes in accounts payable (ΔAP)		-120	-12	-13	-15	-16	176
24	Changes in cash balance (ΔCB)		50	5	6	6	7	-73
25	Changes in working capital		80	8	9	10	11	-117

Working capital
 $= AR + CB - AP$
 $= 15\% + 5\% - 12\% = 8\%$

Working capital (WK in the cash flow)

	A	B	C	D	E	F	G	H
27	Method 1							
28	Year	0	1	2	3	4	5	6
29	Sales		1000	1100	1210	1331	1464	
30	Changes in accounts receivable (ΔAR)		-150	-15	-17	-18	-20	220
31	Total cash receipts		850	1085	1194	1313	1444	220
32	Investment costs	2000						
33	Changes in accounts payable (ΔAP)		-120	-12	-13	-15	-16	176
34	Changes in cash balance (ΔCB)		50	5	6	6	7	-73
35	Operating costs		500	530	562	596	631	
36	Total cash expenditures	2000	430	523	554	587	622	102
37	Net cash flow	-2000	420	562	639	726	822	117
38								
39	Method 2							
40	Year		1	2	3	4	5	6
41	Sales		1000	1100	1210	1331	1464	
42	Total cash receipts		1000	1100	1210	1331	1464	
43	Investment costs	2000						
44	Changes in working capital		80	8	9	10	11	-117
45	Operating costs	0	500	530	561.8	596	631	
46	Total cash expenditures	2000	580	538	571	605	642	-117
47	Net cash flow	-2000	420	562	639	726	822	117
48								
49	Check for consistency (row 37 - row 47)	0	0	0	0	0	0	0



Cash flow statement (Market analysis)

Year	0	1	...	...	n
1. Receipts					
a) Gross sales	●	●	●		
b) ΔAR					
c) Salvage value					
2. Expenditures					
a) Investment costs	●	●	●		
b) Operating costs					
c) ΔAP					
d) ΔCB					
3. Market net cash flow	= (1) – (2)				

Indirect tax-
inclusive
prices

Exclude sunk cost,
but include
opportunity of of
existing assets!

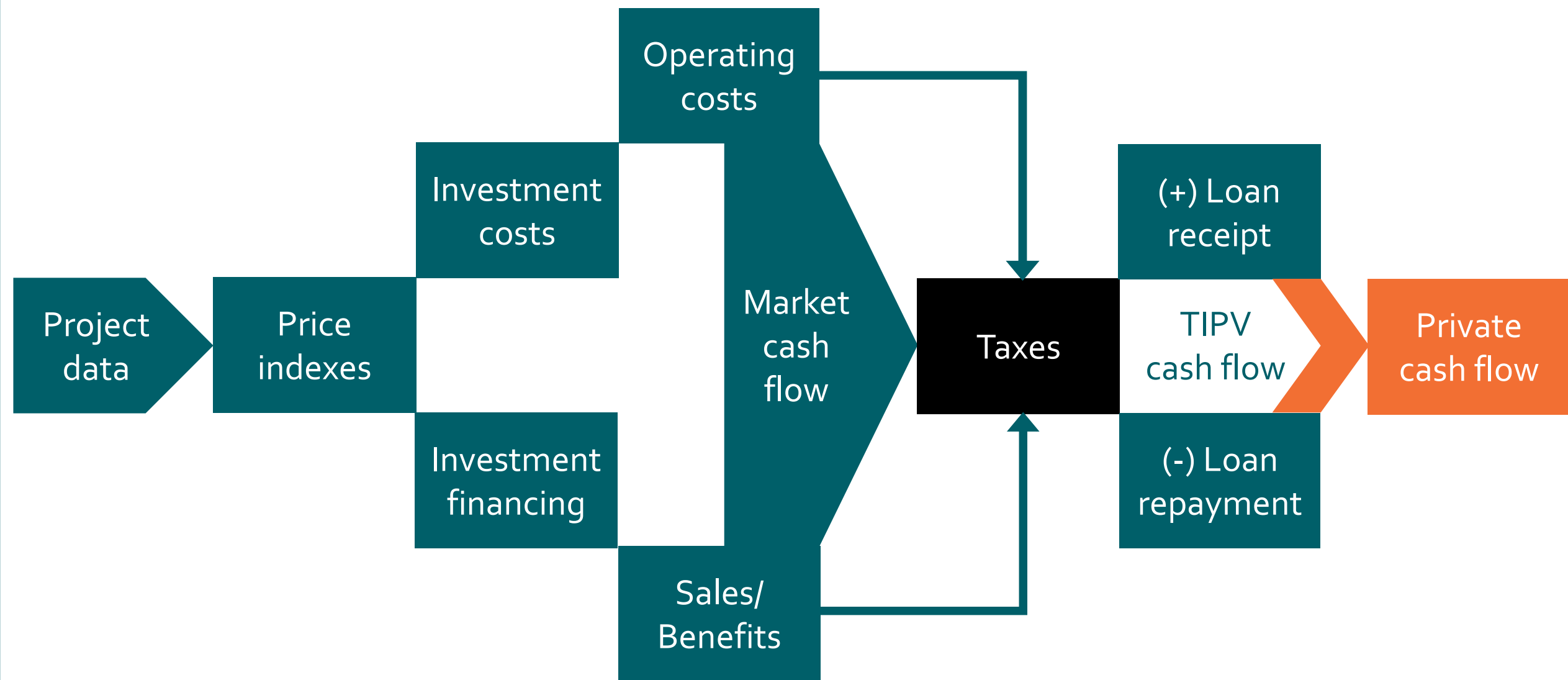
Cash flow statement (Market analysis)

Year	0	1	...	...	n
1. Receipts					
a) Gross sales					
b) Salvage value					
2. Expenditures					
a) Investment costs					
b) Working capital investment					
c) Operating costs					
3. Market net cash flow	= (1) – (2)				

Indirect tax-
inclusive
prices

Include
import duties,
handling fees

Not include
depreciation!



Taxes (Income tax)

Year	1	... n
Net sales (excluding sales/value-added taxes)		
(-) Fixed operating costs	Including depreciation	
(-) Variable operating costs		
EBIT (Earnings before interest and tax)		
(-) Interest expense		
EBT (Earnings before tax)		
(-) Income tax		
Net income (profit)		

Taxes

(Income tax)

Year	1	... n
Net sales		
(-) Direct costs		Including depreciation
Gross profit		
(-) Operating expenses		
EBIT		
(-) Interest expense		
EBT		
(-) Income tax		
Net income (profit)		

Provisions for loss carry forward should be taken into account.

Taxes

(Income tax)

Year	1	... n
Net sales	Cost of sales	
(-) Cost of goods sold (COGS)		Including depreciation
Gross profit		
(-) Operating expenses		
EBIT		
(-) Interest expense		
EBT		
(-) Income tax		
Net income (profit)		

Taxes

(Income tax)

Year	1	... n
Net sales		
(-) Direct costs / COGS		Excluding depreciation
Gross profit		
(-) Operating expenses		
EBITDA		
(-) Depreciation expense		
EBIT		
(-) Interest expense		
EBT		
(-) Income tax		
Net income (profit)		

Income tax (or direct tax) is a cash outflow item from private points of view.

There are three cases:

Read “GST input tax credits”

1) A project's output is levied VAT

- $\Delta\text{VAT} = \text{VAT on output} - \text{VAT on inputs}$ (add additional line item reporting the VAT payments to the government)
- If $\Delta\text{VAT} > 0$, implies a cash outflow (payment); and if $\Delta\text{VAT} < 0$, implies a cash inflow (refund).

2) A project's output is exempt from VAT

- The project will continue to pay VAT on its purchases of inputs (i.e. VAT as a cash outflow). There will not be an additional line item reporting the VAT payments to the government.
- The net effect of the VAT is to increase the cost of the inputs and hence the financial cash outflow of the project.

3) Project's output is exported

- $\Delta\text{VAT} = 0 - \text{VAT on inputs} = -\text{VAT on inputs}$ (add additional line item reporting a refund of VAT paid)
- This should be reported as a negative cost or a cash inflow to the project.

CASE 1: PROJECT'S OUTPUT IS LEVIED VAT

CASE 3: PROJECT'S OUTPUT IS EXPORTED

	B	C	D	E	F	G	H	I	J
48	Year	0	1	2	3	4	5	6	7
49	A. RECEIPTS								
50	Net sales			3000	4000	5500	3800	3000	
51	VAT @10%			300	400	550	380	300	
52	ΔAR @10%			-330	-110	-165	187	88	330
53	Liquidation value								1060
54	Total cash inflow			2970	4290	5885	4367	3388	1390
55	B. EXPENDITURES								
56	a) Investment costs								
57	Materials								
58	- Traded goods	500	500						
59	Tariff @15%	75	75						
60	VAT @10%	58	58						
61	- Nontraded goods	400	300						
62	VAT @10%	40	30						
63	Equipment								
64	- Traded goods (cif)	600	2000						
65	Tariff @15%	90	300						
66	VAT @10%	69	230						
67	b) Operating costs (materials)								
68	- Traded goods (cif)			600	750	800	700	600	
69	Tariff @15%			90	113	120	105	90	
70	VAT @10%			69	86	92	81	69	
71	- Nontraded goods			200	250	300	280	250	
72	VAT @10%			20	25	30	28	25	
73	ΔAP @15%			-33	-8	-8	3	5	41
74	ΔCB @5%			49	12	6	-7	-8	-52
75	c) Construction labour								
76	- Skilled	150	100						
77	- Unskilled	200	250						
78	d) Operating labour								
79	- Skilled			100	150	200	150	125	
80	- Unskilled			50	70	90	80	60	
81	e) Other costs			25	30	35	30	25	
82	VAT @10%			3	3	4	3	3	
83	f) Rehabilitation cost								500
84	Total cash outflow	2182	3843	1172	1481	1668	1452	1243	490
85	C) Tax payments								
86	- VAT	-167	-318	209	286	425	269	204	0
87	- Royalty (@10% on sales)	0	0	300	400	550	380	300	0
88	- Income tax	0	0	141	267	506	246	268	0
89	Total tax payments	-167	-318	650	953	1,481	895	771	0
90	D) Net cash flow (TIPV)	-2015	-3525	1148	1856	2736	2020	1373	900
91									
92	IRR	18.26%							

	B	C	D	E	F	G	H	I	J
48	Year	0	1	2	3	4	5	6	7
49	A. RECEIPTS								
50	Foreign sales			3000	4000	5500	3800	3000	
51	VAT @ 0%			0	0	0	0	0	
52	ΔAR @10%			-300	-100	-150	170	80	300
53	Liquidation value								1060
54	Total cash inflow			2700	3900	5350	3970	3080	1360
55	B. EXPENDITURES								
56	a) Investment costs								
57	Materials								
58	- Traded goods	500	500						
59	Tariff @15%	75	75						
60	VAT @10%	58	58						
61	- Nontraded goods	400	300						
62	VAT @10%	40	30						
63	Equipment								
64	- Traded goods (cif)	600	2000						
65	Tariff @15%	90	300						
66	VAT @10%	69	230						
67	b) Operating costs (materials)								
68	- Traded goods (cif)			600	750	800	700	600	
69	Tariff @15%			90	113	120	105	90	
70	VAT @10%			69	86	92	81	69	
71	- Nontraded goods			200	250	300	280	250	
72	VAT @10%			20	25	30	28	25	
73	ΔAP @15%			-33	-8	-8	3	5	41
74	ΔCB @5%			49	12	6	-7	-8	-52
75	c) Construction labour								
76	- Skilled	150	100						
77	- Unskilled	200	250						
78	d) Operating labour								
79	- Skilled			100	150	200	150	125	
80	- Unskilled			50	70	90	80	60	
81	e) Other costs			25	30	35	30	25	
82	VAT @10%			3	3	4	3	3	
83	f) Rehabilitation cost								500
84	Total cash outflow	2182	3843	1172	1481	1668	1452	1243	490
85	C) Tax payments								
86	- VAT	-167	-318	-92	-114	-126	-112	-97	0
87	- Royalty (@10% on sales)	0	0	300	400	550	380	300	0
88	- Income tax	0	0	141	267	506	246	268	0
89	Total tax payments	-167	-318	350	553	931	515	471	0
90	D) Net cash flow (TIPV)	-2015	-3525	1178	1866	2751	2003	1365	870
91									
92	IRR	18.37%							

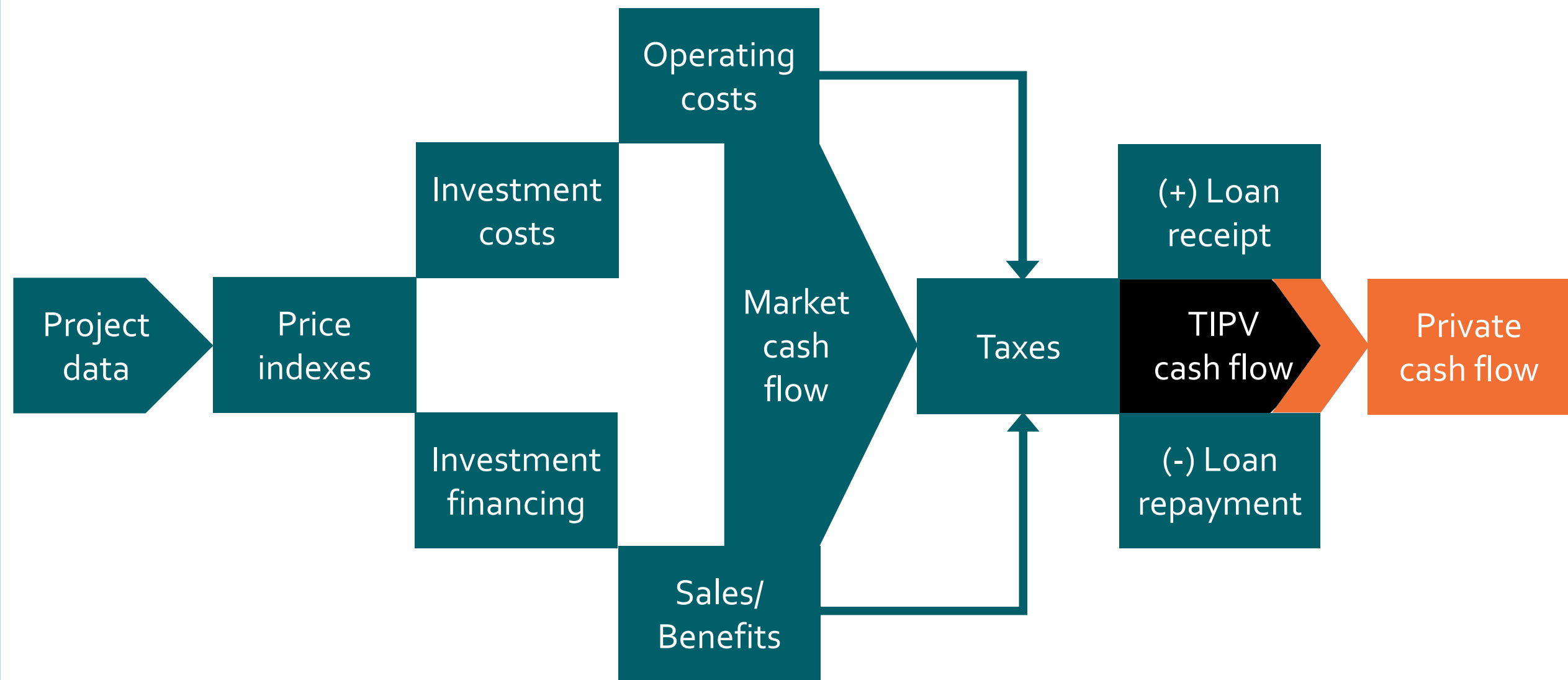
Taxes

(Case 2: Output is exempt from VAT)

	B	C	D	E	F	G	H	I	J
48	Year	0	1	2	3	4	5	6	7
49	A. RECEIPTS								
50	Gross sales = Net sales			3000	4000	5500	3800	3000	
51	VAT @ 0%			0	0	0	0	0	
52	ΔAR @10%			-300	-100	-150	170	80	300
53	Liquidation value								1060
54	Total cash inflow			2700	3900	5350	3970	3080	1360
55	B. EXPENDITURES								
56	a) Investment costs								
57	Materials								
58	- Traded goods	500	500						
59	Tariff @15%	75	75						
60	VAT @10%	58	58						
61	- Nontraded goods	400	300						
62	VAT @10%	40	30						
63	Equipment								
64	- Traded goods (cif)	600	2000						
65	Tariff @15%	90	300						
66	VAT @10%	69	230						
67	b) Operating costs (materials)								
68	- Traded goods (cif)			600	750	800	700	600	
69	Tariff @15%			90	113	120	105	90	
70	VAT @10%			69	86	92	81	69	
71	- Nontraded goods			200	250	300	280	250	
72	VAT @10%			20	25	30	28	25	
73	ΔAP @15%			-33	-8	-8	3	5	41
74	ΔCB @5%			49	12	6	-7	-8	-52
75	c) Construction labour								
76	- Skilled	150	100						
77	- Unskilled	200	250						
78	d) Operating labour								
79	- Skilled			100	150	200	150	125	
80	- Unskilled			50	70	90	80	60	
81	e) Other costs			25	30	35	30	25	
82	VAT @10%			3	3	4	3	3	
83	f) Rehabilitation cost								500
84	Total cash outflow	2182	3843	1172	1481	1668	1452	1243	490
85	C) Tax payments								
86	- VAT	0	0	0	0	0	0	0	0
87	- Royalty (@10% on sales)	0	0	300	400	550	380	300	0
88	- Income tax	0	0	141	267	506	246	268	0
89	Total tax payments	0	0	441	667	1,056	626	568	0
90	D) Net cash flow (TIPV)	-2182	-3843	1086	1752	2626	1891	1269	870
91									
92	IRR	13.58%							

Cash outflow includes VAT on purchases of inputs. VAT on inputs are cash expenditures.

This line item should be removed.



Cash flow statement

(Market analysis – TIPV)

Year	0	1	...	...	n
1. Receipts					
a) Net sales					
b) VAT on sales					
c) ΔAR					
d) Salvage value					
2. Expenditures					
a) Investment costs					
b) Operating costs					
c) VAT on purchases					
d) ΔAP					
e) ΔCB					
3. Market net cash flow					
4. Tax payments					
a) VAT (Payment, Refund) = 1b) – 2c)					
b) Income tax					
5. Market net cash flow (TIPV)					

Indirect tax
exclusive
prices

In case of tax
exemption, this
line item is
omitted

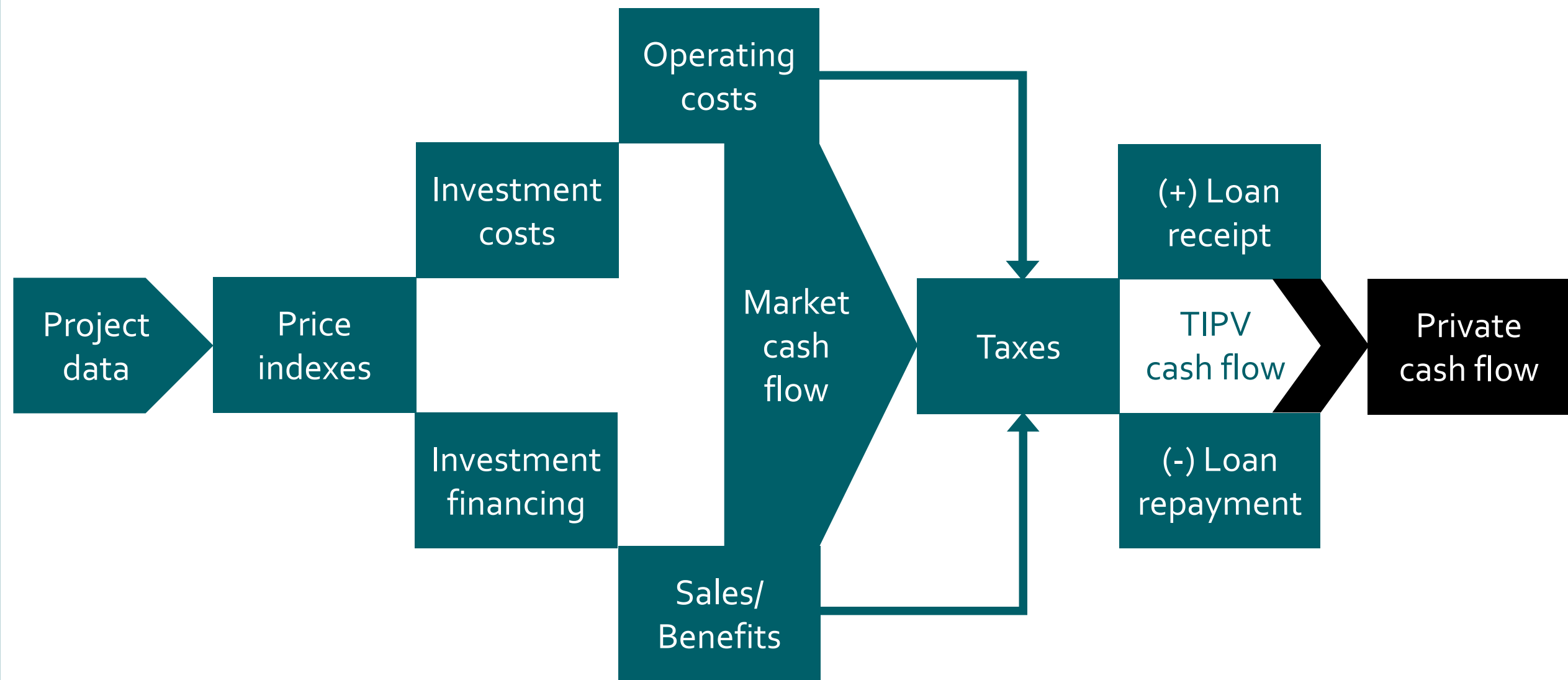
$$= (1) - (2)$$

$$= (3) - (4)$$

Cash flow statement (Market analysis – TIPV)

Year	0	1	...	...	n
1. Receipts					
a) Net sales					
b) VAT on sales					
c) Salvage value					
2. Expenditures					
a) Investment costs					
b) Operating costs					
c) Changes in working capital					
d) VAT on purchases					
3. Market net cash flow	= (1) – (2)				
4. Tax payments					
a) VAT (Payment, Refund) = 1b) – 2d)					
b) Income tax					
5. Market net cash flow (TIPV)	= (3) – (4)				

In cases of foreign sales and tax exemption, this line item is zero



Cash flow statement (Private analysis – EPV)

Year	0	1	...	...	n
1. Market net cash flow (TIPV)					
2. Loan receipts	+				
3. Interest and principal repayments		+	+	+	
4. Private net cash flow (EPV)	= (1) + (2) – (3)				

Year	0	1	...	...	n
1. Market net cash flow (TIPV)					
2. Debt finance flow	+	-	-	-	
3. Private net cash flow (EPV)	= (1) + (2)				

Financial cash flows (Discussion)

- The market cash flow which shows the financial flows expected to result from the project itself and contains no information about the way the project might be financed.
- Different loan terms will generate different profiles of financing inflows and outflows for the project's owners.
- We are able to consider any given project's profitability independently of the terms on which its debt finance is arranged by using **market appraisal**.
- If we were to ignore this issue, it is conceivable that a "bad" project could be made to look good; and that a "good" project may look "bad".

Financial cash flow of a project

Incremental cash flows



Incremental cash flows (Explanation)

- The use of incremental cash flow technique is to evaluate investments that aim at the improvement of existing schemes.
- For examples, replacement of an outdated piece of machinery by a more modern model, a rehabilitation of a factory, an irrigation project.
- It is assumed that there is already a cash flow from the existing project and that the main objective of a proposed additional investment is to improve the net cash flow, either by decreasing cost, or by increasing benefits, or by both of these at the same time.
- It is the difference between the net benefit flow with the new investment and the net benefit flow without this investment.
- However, there are pitfalls in appraising or evaluating investments on the basis of incremental cash flows.

Incremental cash flows (Examples)

EXAMPLE 1

	0	1	2	3	–	10
1 Net Cash Flow With rehabilitation	–1500	750	750	750	–	750
2 Net Cash Flow Without rehabilitation		500	500	500	–	500
3 Incremental Cash Flow (1–2)	–1500	250	250	250	–	250

EXAMPLE 2

1 Net Cash Flow With rehabilitation	–1500	–250	–250	–250	–	–250
2 Net Cash Flow Without rehabilitation		–500	–500	–500	–	–500
3 Incremental Cash Flow (1–2)	–1500	250	250	250	–	250

EXAMPLE 3

1 Net Cash Flow With closure	0	0	0	0	–	0
2 Net Cash Flow Without closure		–500	–500	–500		–500
3 Incremental Cash Flow (1–2)		500	500	500	–	500

Financial cash flow of a project

An illustrative example



Step 1: Setting up the "Key Variables" table

Table 1: Key Variables		Project			Project				
Investment Costs		No.	Price	Cost	Operating Costs		No.	Price	Cost
(i) Fixed Investment		(units)	(\$)		Rent on land (Ha)		100	30	3,000
Farm equipment (units)		4	100,000	400,000	Fuel (litres)		2,500	0.7	1,750
Vehicles (units)		3	30,000	90,000	Seeds (Kg)		250	20	5,000
Buildings (m2)		250	1,000	250,000	Fertilizers (tonnes)		3	500	1,500
TOTAL				\$740,000	Insecticides (litres)		3,000	30	90,000
(ii) Working Capital					Water (ML)		900	20	18,000
Fertilizer Stocks (tons)		2	500	1,000	Spares		12	1,000	12,000
Insecticide stocks (Litres)		2500	30	75,000	Casual labour (days)		100	60	6,000
Equipment spare parts (units)		10	1,000	10,000	Administration (/month)		12	1000	12,000
Fuel stocks (Litres)		500	0.7	350	Insurance (p.a.)		1%	826,350	8,263
TOTAL				\$86,350.0	Management (/month)		12	3,000	36,000
(iii) Salvage Value		10%		\$74,000	Miscellaneous		1	7,700	7,700
					TOTAL (Market prices)				\$201,213
Depreciation		Life(yrs)	Amount p.a.		Revenues				
	Equipmnt.	10	40,000		Apples (tons)		100	1,000	100,000
	Vehicles	5	18,000		Peaches (tons)		90	1,250	112,500
	Buildings	20	12,500		Pears (tons)		75	1,500	112,500
Financing		Amount	Interest	Life (yrs)	TOTAL				\$325,000
Loan		\$700,000	3.5%	10			Conversion factor		1000
Overdraft		\$40,000	5.0%	4	Capacity Output				
Discount rate =		5.0%	10.0%	15.0%		2016	2017	2018	2019+
Tax rate on profits =		25.0%			%	25%	50%	75%	100%

Step 2: Setting up the project's cash flow for the Market Analysis

TABLE 2: MARKET NET CASH FLOW												
	2015	2016	2017	2018	2019	2020	2030	2031	2032	2033	2034	2035
ITEM/YEAR	0	1	2	3	4	5	15	16	17	18	19	20
Investment costs												
Fixed Investment	-740.0											74.0
Working Capital		-86.4										86.4
Total Investment	-740.0	-86.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	160.3
Operating Costs		-50.3	-100.6	-150.9	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2
Revenues		81.3	162.5	243.8	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0
Net Cash Flow	-740.0	-55.4	61.9	92.8	123.8	123.8	123.8	123.8	123.8	123.8	123.8	284.1
(Before Financing & Tax)												
	5%	10%	15%									
Net Present Value =	609.6	100.4	-178.3									
IRR =	11.5%											

Step 3: Setting up the cash flow for the financing of the project

TABLE 3: PRIVATE NET CASH FLOW																	
ITEM/YEAR	2015	2016	2017	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
	0	1	2	7	8	9	10	11	12	13	14	15	16	17	18	19	20
(i) Financing																	
Principal																	
Loan	-700.0	59.7	61.8	73.3	75.9	78.6	81.3										
Overdraft		-40.0															
Interest																	
Loan		24.5	22.4	10.8	8.3	5.6	2.8										
Overdraft			2.0														
Net Financing Flows	-700.0	44.2	86.2	84.2	84.2	84.2	84.2										
NCF(equity, pre-tax)	-40.0	-99.6	-24.3	39.6	39.6	39.6	39.6	123.8	123.8	123.8	123.8	123.8	123.8	123.8	123.8	123.8	284.1
(ii) Taxes																	
Revenues		81.3	162.5	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0
Operating Costs		-50.3	-100.6	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2
Depreciation		-70.5	-70.5	-52.5	-52.5	-52.5	-52.5	-12.5	-12.5	-12.5	-12.5	-12.5	-12.5	-12.5	-12.5	-12.5	-12.5
Interest on loans		-24.5	-24.4	-10.8	-8.3	-5.6	-2.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Profits (before tax)		-64.1	-33.0	60.5	63.0	65.7	68.4	111.3	111.3	111.3	111.3	111.3	111.3	111.3	111.3	111.3	111.3
Taxes Liabe		-16.0	-8.3	15.1	15.8	16.4	17.1	27.8	27.8	27.8	27.8	27.8	27.8	27.8	27.8	27.8	27.8
(iii) Equity (after tax)																	
Private Cash Flow	-40.0	-83.6	-16.0	24.5	23.9	23.2	22.5	96.0	96.0	96.0	96.0	96.0	96.0	96.0	96.0	96.0	256.3
	5%	10%	15%														
NPV =	\$483.3	\$196.4	\$59.6														
IRR =	19.1%																

Use the "PPMT" and "IPMT" formulae to calculate debt payments

Depreciation and interest payments are included in the calculation of taxable profits

The private cash flow is derived by subtracting debt financing flows and profits taxes from the market net cash flow in Table 2

Use the "PPMT" and "IPMT" formulae to calculate debt payments

Depreciation and interest payments are included in the calculation of taxable profits

The private cash flow is derived by subtracting debt financing flows and profits taxes from the market net cash flow in Table 2

Step 4: Calculating business income taxes and after-tax flows

TABLE 3: PRIVATE NET CASH FLOW																	
ITEM/YEAR	2015	2016	2017	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
	0	1	2	7	8	9	10	11	12	13	14	15	16	17	18	19	20
(i) Financing																	
Principal																	
Loan	-700.0	59.7	61.8	73.3	75.9	78.6	81.3										
Overdraft		-40.0															
Interest																	
Loan		24.5	22.4	10.8	8.3	5.6	2.8										
Overdraft			2.0														
Net Financing Flows	-700.0	44.2	86.2	84.2	84.2	84.2	84.2										
NCF(equity, pre-tax)	-40.0	-99.6	-24.3	39.6	39.6	39.6	39.6	123.8	123.8	123.8	123.8	123.8	123.8	123.8	123.8	123.8	284.1
(ii) Taxes																	
Revenues		81.3	162.5	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0
Operating Costs		-50.3	-100.6	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2
Depreciation		-70.5	-70.5	-52.5	-52.5	-52.5	-52.5	-12.5	-12.5	-12.5	-12.5	-12.5	-12.5	-12.5	-12.5	-12.5	-12.5
Interest on loans		-24.5	-24.4	-10.8	-8.3	-5.6	-2.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Profits (before tax)		-64.1	-33.0	60.5	63.0	65.7	68.4	111.3	111.3	111.3	111.3	111.3	111.3	111.3	111.3	111.3	111.3
Taxes Liabe		-16.0	-8.3	15.1	15.8	16.4	17.1	27.8	27.8	27.8	27.8	27.8	27.8	27.8	27.8	27.8	27.8
(iii) Equity (after tax)																	
Private Cash Flow	-40.0	-83.6	-16.0	24.5	23.9	23.2	22.5	96.0	96.0	96.0	96.0	96.0	96.0	96.0	96.0	96.0	256.3
	5%	10%	15%														
NPV =	\$483.3	\$196.4	\$59.6														
IRR =	19.1%																

Use the "PPMT" and "IPMT" formulae to calculate debt payments

Depreciation and interest payments are included in the calculation of taxable profits

The private cash flow is derived by subtracting debt financing flows and profits taxes from the market net cash flow in Table 2

Use the "PPMT" and "IPMT" formulae to calculate debt payments

Depreciation and interest payments are included in the calculation of taxable profits

The private cash flow is derived by subtracting debt financing flows and profits taxes from the market net cash flow in Table 2

Step 5: Deriving the Private (after-tax) net cash flow

TABLE 3: PRIVATE NET CASH FLOW

ITEM/YEAR	2015 0	2016 1	2017 2	2022 7	2023 8	2024 9	2025 10	2026 11	2027 12	2028 13	2029 14	2030 15	2031 16	2032 17	2033 18	2034 19	2035 20
Use the "PPMT" and "IPMT" formulae to calculate debt payments																	
Depreciation and interest payments are included in the calculation of taxable profits																	
(i) Financing																	
Principal																	
Loan	-700.0	59.7	61.8	73.3	75.9	78.6	81.3										
Overdraft		-40.0															
Interest																	
Loan		24.5	22.4	10.8	8.3	5.6	2.8										
Overdraft			2.0														
Net Financing Flows	-700.0	44.2	86.2	84.2	84.2	84.2	84.2										
NCF(equity, pre-tax)	-40.0	-99.6	-24.3	39.6	39.6	39.6	39.6	123.8	123.8	123.8	123.8	123.8	123.8	123.8	123.8	123.8	284.1
(ii) Taxes																	
Revenues		81.3	162.5	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0	325.0
Operating Costs		-50.3	-100.6	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2	-201.2
Depreciation		-70.5	-70.5	-52.5	-52.5	-52.5	-52.5	-12.5	-12.5	-12.5	-12.5	-12.5	-12.5	-12.5	-12.5	-12.5	-12.5
Interest on loans		-24.5	-24.4	-10.8	-8.3	-5.6	-2.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Profits (before tax)		-64.1	-33.0	60.5	63.0	65.7	68.4	111.3	111.3	111.3	111.3	111.3	111.3	111.3	111.3	111.3	111.3
Taxes Liabe		-16.0	-8.3	15.1	15.8	16.4	17.1	27.8	27.8	27.8	27.8	27.8	27.8	27.8	27.8	27.8	27.8
(iii) Equity (after tax)																	
Private Cash Flow	-40.0	-83.6	-16.0	24.5	23.9	23.2	22.5	96.0	96.0	96.0	96.0	96.0	96.0	96.0	96.0	96.0	256.3
The private cash flow is derived by subtracting debt financing flows and profits taxes from the market net cash flow in Table 2																	
	5%	10%	15%														
NPV =	\$483.3	\$196.4	\$59.6														
IRR =	19.1%																

Financial cash flow of a project

Q&A section

